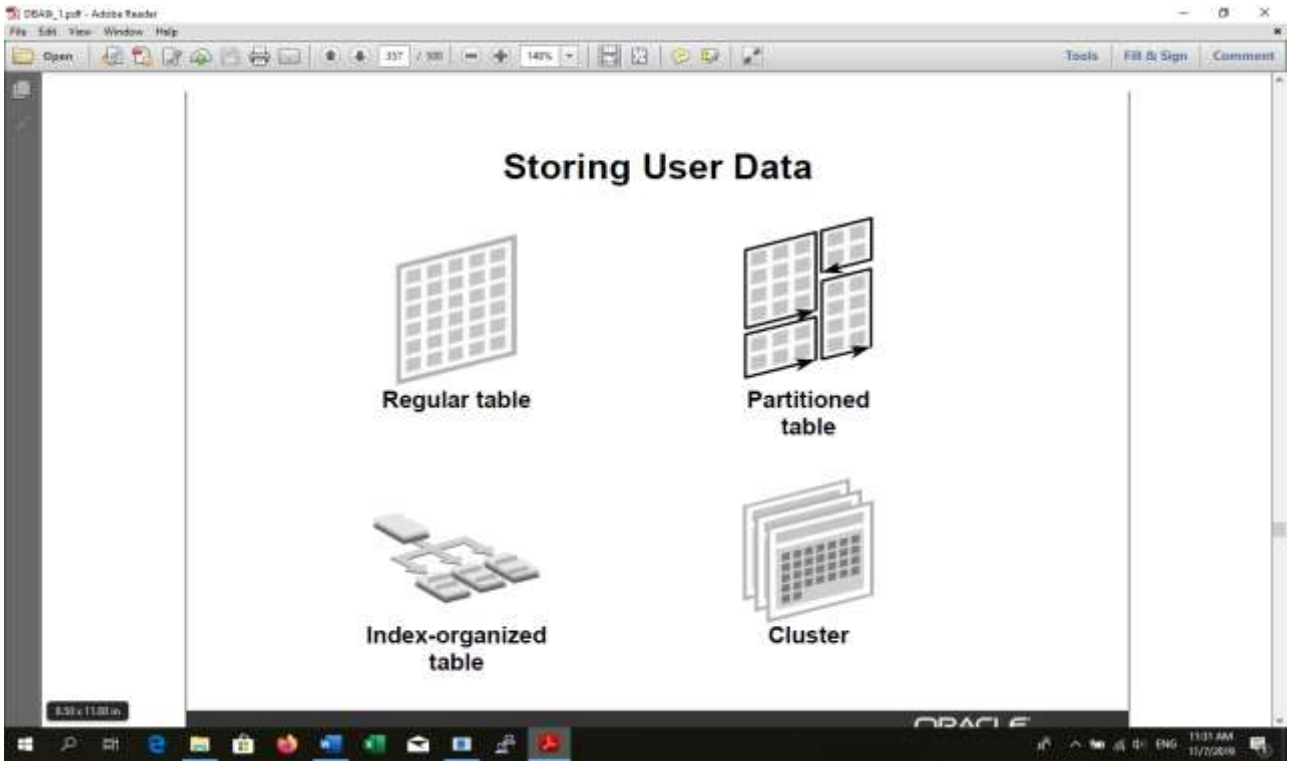
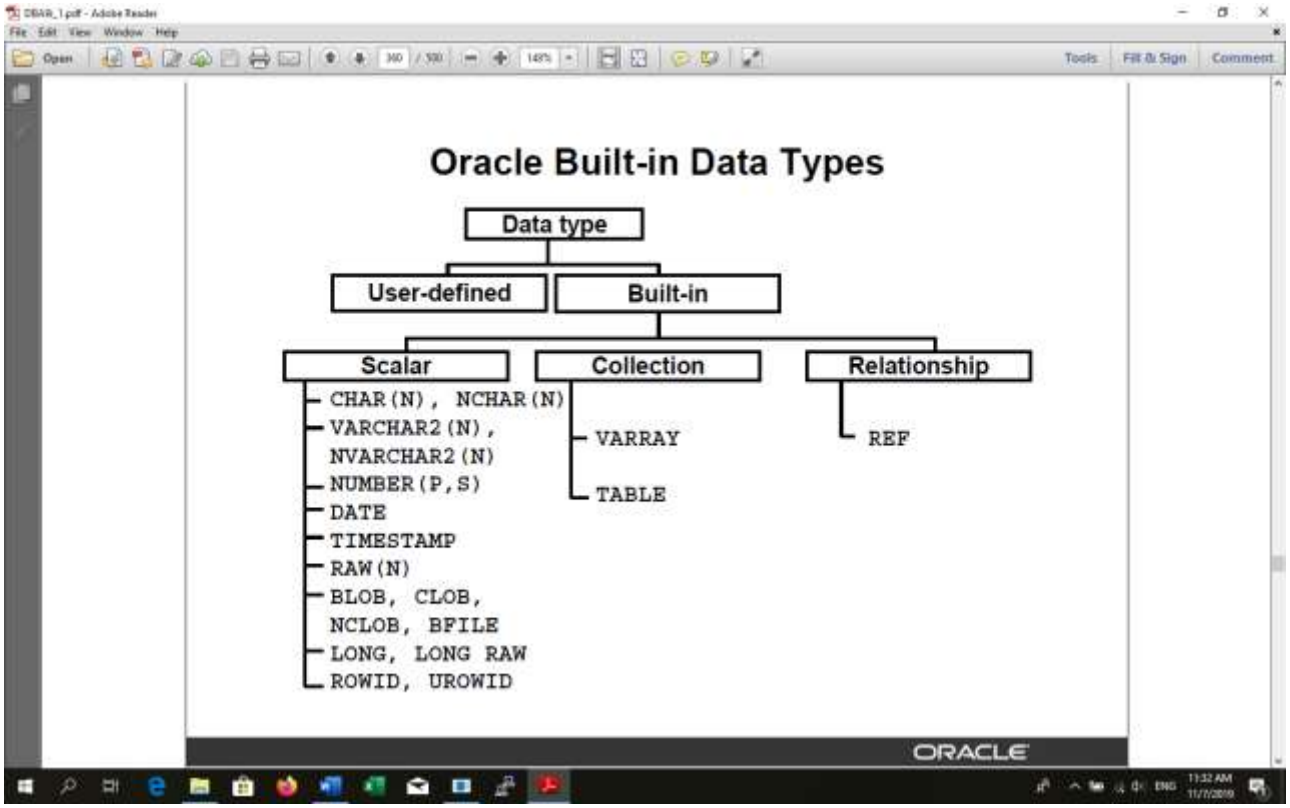
Tabele de date si indecsi

A. Tabele de date





ROWID Format

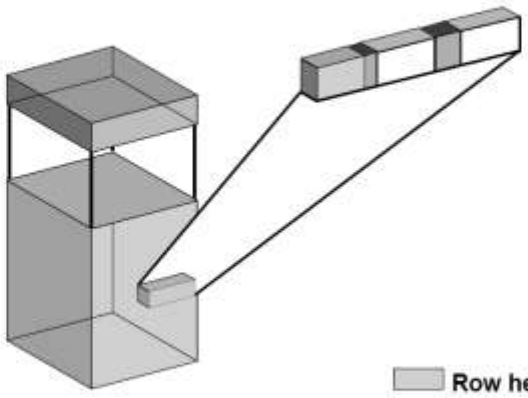
- Extended ROWID Format

OOOOOO	FFF	BBBBBB	RRR
Data object number	Relative file number	Block number	Row number

- Restricted ROWID Format

BBBBBBBB	.	RRRR	.	FFFF
Block number		Row number		File number

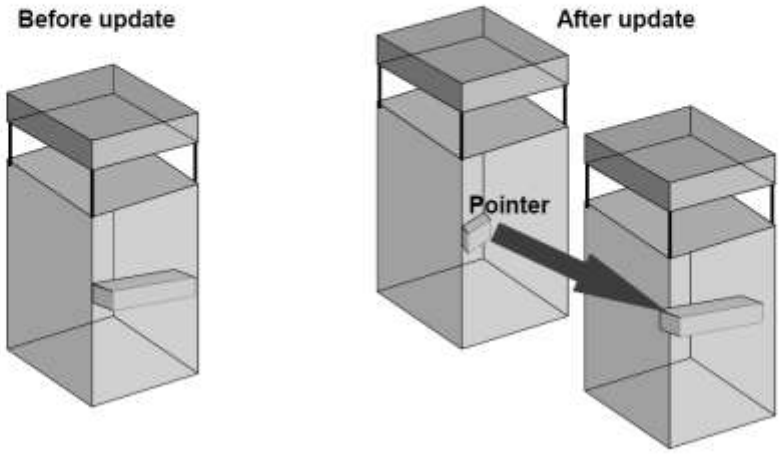
Structure of a Row



Database block

Row header
Column length
Column value

Row Migration and Chaining



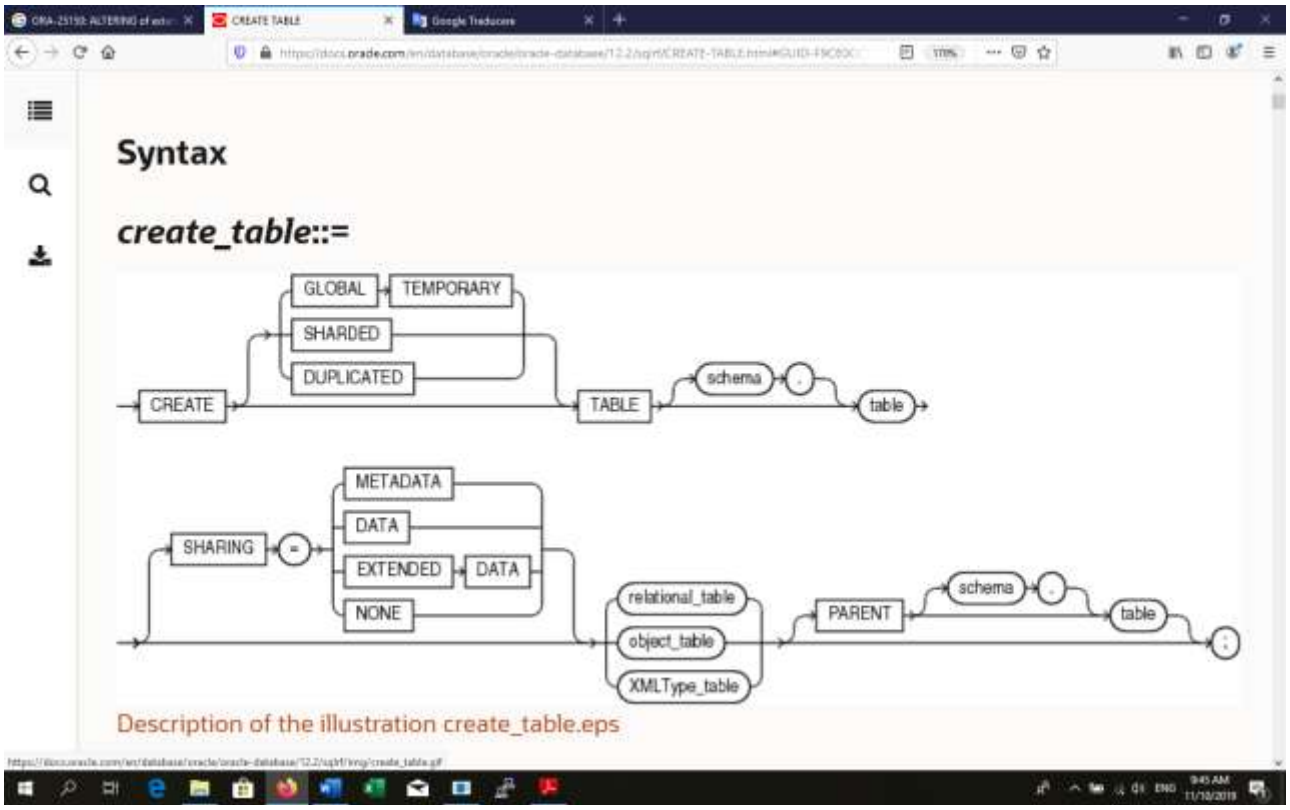
ORACLE

Obtaining Table Information

Information about tables can be obtained by querying the following views:

- DBA_TABLES
- DBA_OBJECTS

ORACLE



1) Crearea unei tabele permanente:

```
SQL> CREATE TABLE studenti
        (facultate char(30)  DEFAULT  'Automatica si Calculatoare',
         catedra char(20),
        cnp number(13),
        nume     varchar2(30),
        data_nastere date,
        an_univ number(4)  DEFAULT  2006,
         media_admitere number(5,2) ,
        discip_oblig  varchar2(20) DEFAULT 'Matematica',
        discip_opt     varchar2(20) DEFAULT 'Fizica' ,
        operator  varchar2(20) DEFAULT  user,
        data_op   date DEFAULT sysdate )
        TABLESPACE ubd_data,
        STORAGE (initial 8M),
        PCTUSED 30,
        PCTFREE 20,
        NOLOGGING;
```

2) Crearea unei tabele permanente partitionate:

```
CREATE TABLE angajati_dep
  (
  depart number(2),
  ecuson number(5),
  nume varchar2(30),
  job  varchar2(20),
  data_ang date,
  salariu number(6)
  )
PARTITION BY LIST (depart)
 (
 PARTITION Dep10 VALUES (10),
 PARTITION Dep20 VALUES (20),
 PARTITION Dep30 VALUES (30)
 );
```

SQL> insert into angajati_dep select deptno, empno, ename, job, hiredate, sal from emp;

SQL> select * from angajati_dep partition (dep10);

DEPART	ECUSON	NUME	JOB	DATA_ANG	SALARIU
10	7782	CLARK	MANAGER	09-JUN-81	2450
10	7839	KING	PRESIDENT	17-NOV-81	5000
10	7934	MILLER	CLERK	23-JAN-82	1300

3) Crearea unei tabele temporare de catre userul curent:

a) SQL> create global temporary table emp_temp **on commit delete rows** as
select * from emp;
table created.

Observatie: Datele se pastreaza in tabela doar pe perioada **tranzactiei**, dar tabela ramane creata si in sesiunile urmatoare.

SQL> select * from emp_temp;
no rows selected

SQL> insert into emp_temp select * from emp;
14 rows created.

SQL> select * from emp_temp;
14 rows selected

SQL>commit;

SQL> select * from emp_temp;
no rows selected

b) SQL> create global temporary table emp_temp **on commit preserve rows** as
select * from emp;

Observatie: Datele se pastreaza in tabela doar pe perioada **sesiunii curente**, dar tabela ramane creata si in sesiunile urmatoare.

SQL> select count(*) from emp_temp;

COUNT(*)

14

4) Vizualizarea ROWID-rilor pentru fiecare linie din tabela:

SQL> create table emp_copy as select * from scott.emp;

SQL> select rowid, empno, ename from emp_copy;

ROWID	EMPNO	ENAME
AAAWc2AAHAAAACXAAA	7369	SMITH
AAAWc2AAHAAAACXAAB	7499	ALLEN
AAAWc2AAHAAAACXAAC	7521	WARD
AAAWc2AAHAAAACXAAD	7566	JONES
AAAWc2AAHAAAACXAAE	7654	MARTIN
AAAWc2AAHAAAACXAAF	7698	BLAKE
AAAWc2AAHAAAACXAAG	7782	CLARK
AAAWc2AAHAAAACXAAH	7788	SCOTT
AAAWc2AAHAAAACXAAI	7839	KING
AAAWc2AAHAAAACXAAJ	7844	TURNER
AAAWc2AAHAAAACXAAK	7876	ADAMS
AAAWc2AAHAAAACXAAL	7900	JAMES
AAAWc2AAHAAAACXAAM	7902	FORD
AAAWc2AAHAAAACXAAN	7934	MILLER

5) Vizualizarea ROWNUM-rilor pentru fiecare linie din tabela:

SQL> select rownum, empno, ename from emp_copy where rownum<10;

ROWNUM	EMPNO	ENAME

1	7369	SMITH
2	7499	ALLEN
3	7521	WARD
4	7566	JONES
5	7654	MARTIN
6	7698	BLAKE
7	7782	CLARK
8	7788	SCOTT
9	7839	KING

6) Alocarea unei extensii la o tabela:

```
SQL> alter tablespace bd_data
        add datafile 'e:/temp/abd_data1.dbf'  size 2M;
Tablespace altered.

SQL>  alter table emp_copy
        allocate extent(size 500k datafile  'e:/temp/abd_data1.dbf');
Table altered.
```

7) Stergerea unei coloane dintr-o tabela:

```
SQL>  alter table emp_copy
        drop column comm
        cascade constraints checkpoint 1000;
```

8) Redenumirea unei coloane dintr-o tablela:

```
SQL>  alter table emp_copy
        rename column sal to salary;
```

9) Dezactivarea unei coloane dintr-o tabela:

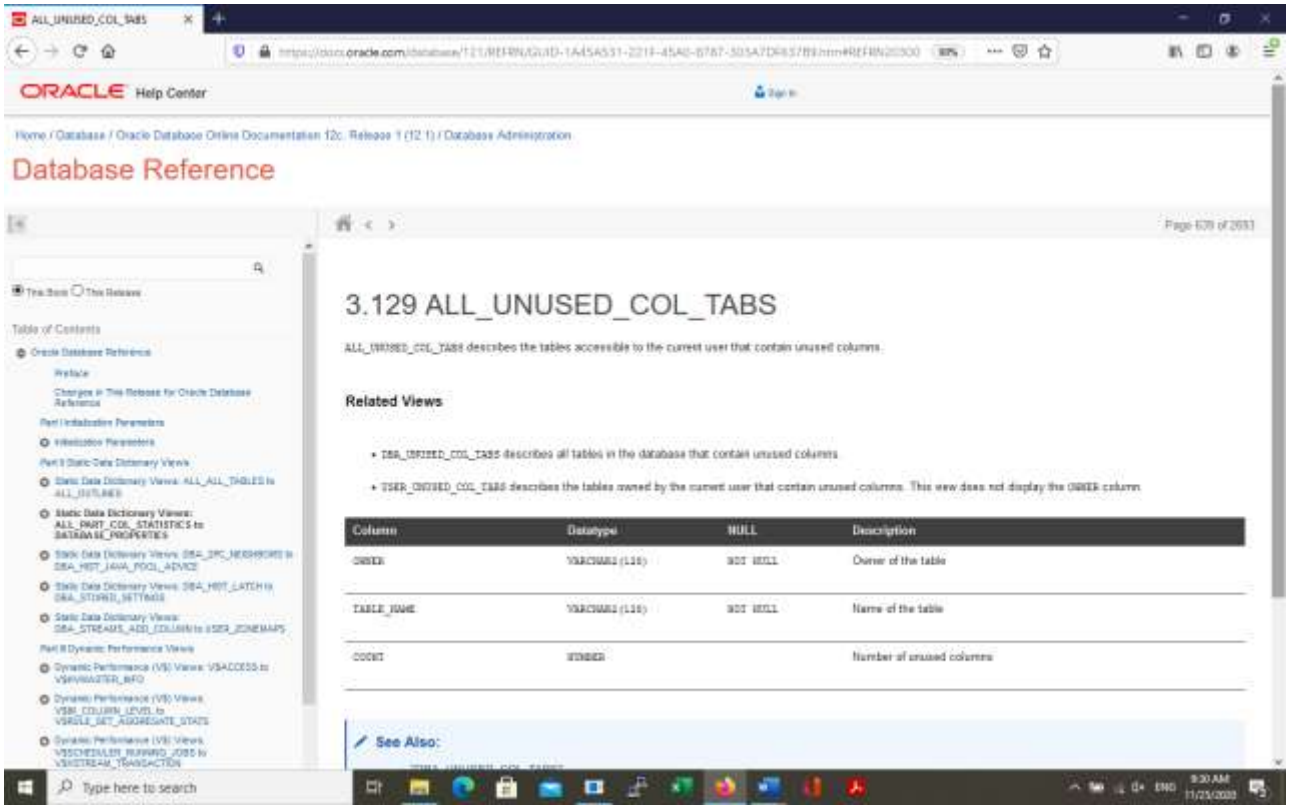
```
SQL>  alter table emp_copy
        set unused column salary
        cascade constraints;
```

10) Informatii despre tabele cu coloane dezactivate:

```
SQL> desc dba_unused_col_tabs;

 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------
 OWNER                                     NOT NULL VARCHAR2(30)
 TABLE_NAME                                NOT NULL VARCHAR2(30)
```

COUNT NUMBER



SQL> select * from dba_unused_col_tabs;

OWNER	TABLE_NAME	COUNT
ABD1	EMP_COPY	1

11) Stergerea din dictionar a coloanelor dezactivate dintr-o tabela:

```
SQL>  alter table emp_copy
        drop unused columns checkpoint 1000;

SQL> select * from dba_unused_col_tabs;
no rows selected
```

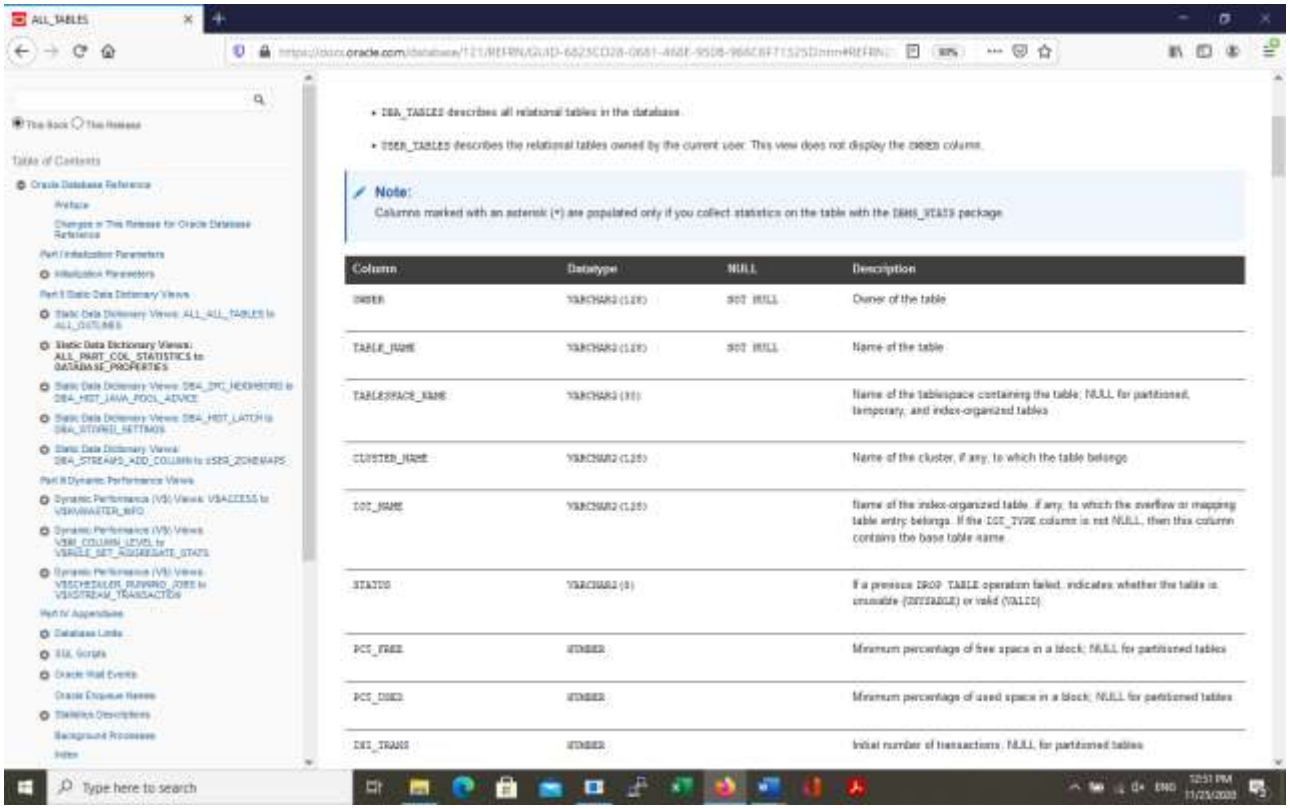
12) Informatii despre toate tabelele din baza de date :

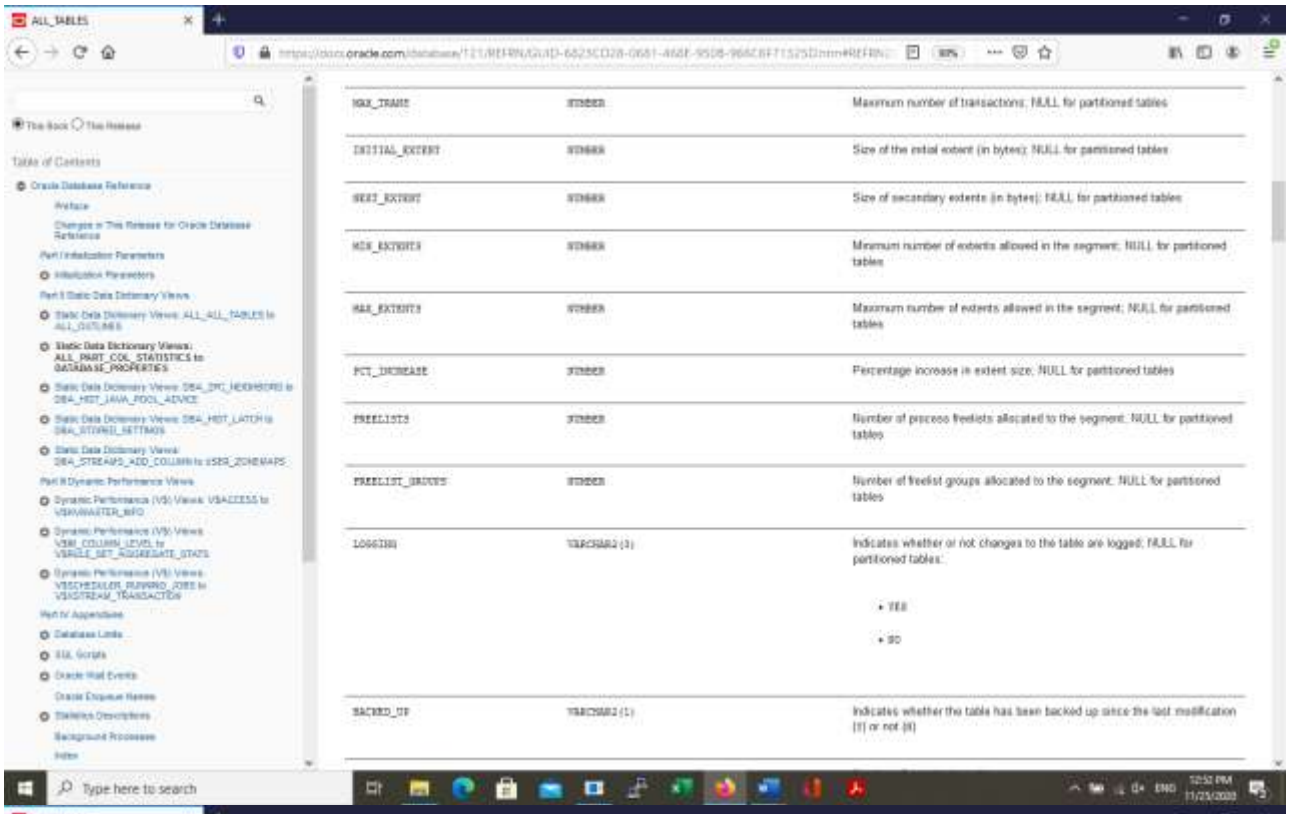
SQL> desc dba_tables;

Nume	Nul?	Type
OWNER	NOT NULL	VARCHAR2(30)
TABLE_NAME	NOT NULL	VARCHAR2(30)
TABLESPACE_NAME		VARCHAR2(30)

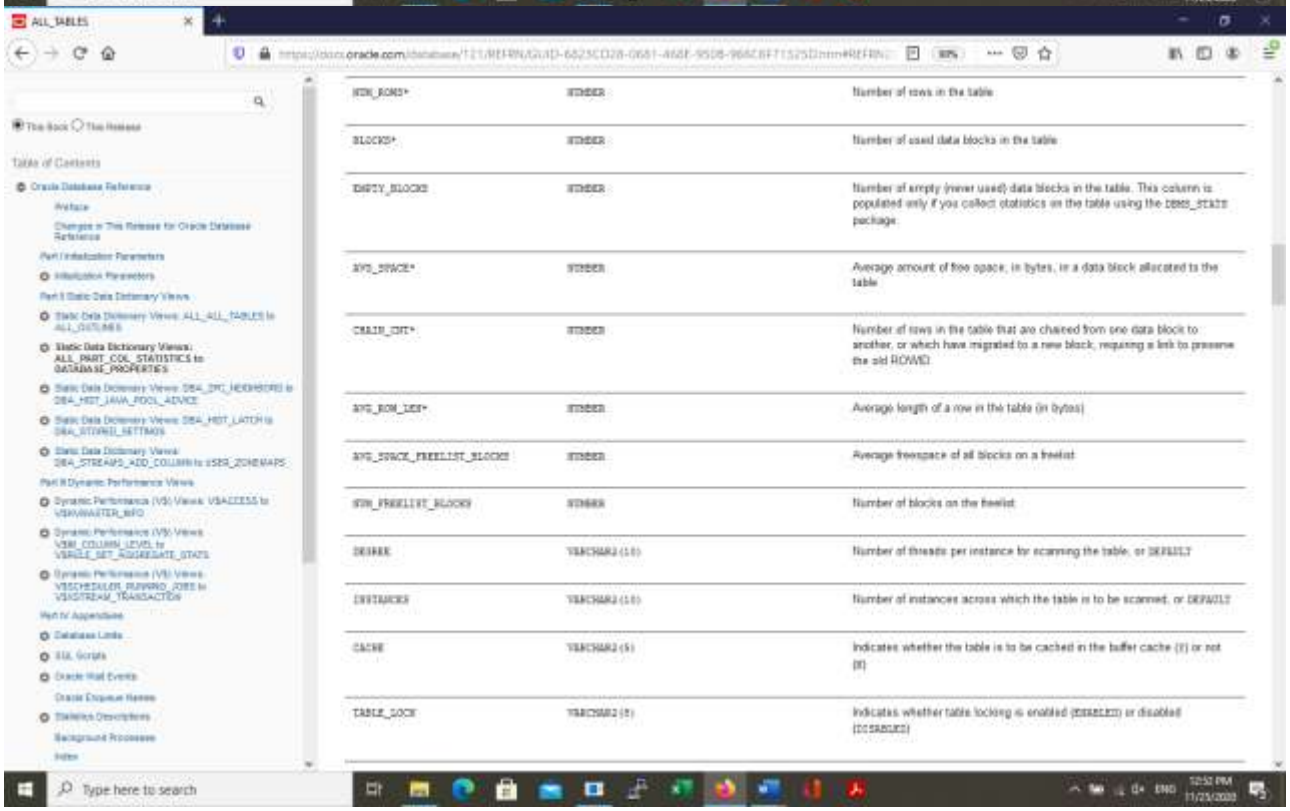
CLUSTER_NAME	VARCHAR2(30)
IOT_NAME	VARCHAR2(30)
PCT_FREE	NUMBER
PCT_USED	NUMBER
INI_TRANS	NUMBER
MAX_TRANS	NUMBER
INITIAL_EXTENT	NUMBER
NEXT_EXTENT	NUMBER
MIN_EXTENTS	NUMBER
MAX_EXTENTS	NUMBER
PCT_INCREASE	NUMBER
FREELISTS	NUMBER
FREELIST_GROUPS	NUMBER
LOGGING	VARCHAR2(3)
BACKED_UP	VARCHAR2(1)
NUM_ROWS	NUMBER
BLOCKS	NUMBER
EMPTY_BLOCKS	NUMBER
AVG_SPACE	NUMBER
CHAIN_CNT	NUMBER
AVG_ROW_LEN	NUMBER
AVG_SPACE_FREELIST_BLOCKS	NUMBER
NUM_FREELIST_BLOCKS	NUMBER
DEGREE	VARCHAR2(10)
INSTANCES	VARCHAR2(10)
CACHE	VARCHAR2(5)
TABLE_LOCK	VARCHAR2(8)
SAMPLE_SIZE	NUMBER
LAST_ANALYZED	DATE
PARTITIONED	VARCHAR2(3)
IOT_TYPE	VARCHAR2(12)
TEMPORARY	VARCHAR2(1)
SECONDARY	VARCHAR2(1)
NESTED	VARCHAR2(3)
BUFFER_POOL	VARCHAR2(7)
ROW_MOVEMENT	VARCHAR2(8)
GLOBAL_STATS	VARCHAR2(3)
USER_STATS	VARCHAR2(3)
DURATION	VARCHAR2(15)
SKIP_CORRUPT	VARCHAR2(8)
MONITORING	VARCHAR2(3)
CLUSTER_OWNER	VARCHAR2(30)
DEPENDENCIES	VARCHAR2(8)
COMPRESSION	VARCHAR2(8)
COMPRESS_FOR	VARCHAR2(30)
DROPPED	VARCHAR2(3)

READ_ONLY	VARCHAR2(3)
SEGMENT_CREATED	VARCHAR2(3)
RESULT_CACHE	VARCHAR2(7)
CLUSTERING	VARCHAR2(3)
ACTIVITY_TRACKING	VARCHAR2(23)
DML_TIMESTAMP	VARCHAR2(25)
HAS_IDENTITY	VARCHAR2(3)
CONTAINER_DATA	VARCHAR2(3)
INMEMORY	VARCHAR2(8)
INMEMORY_PRIORITY	VARCHAR2(8)
INMEMORY_DISTRIBUTE	VARCHAR2(15)
INMEMORY_COMPRESSION	VARCHAR2(17)
INMEMORY_DUPLICATE	VARCHAR2(13)

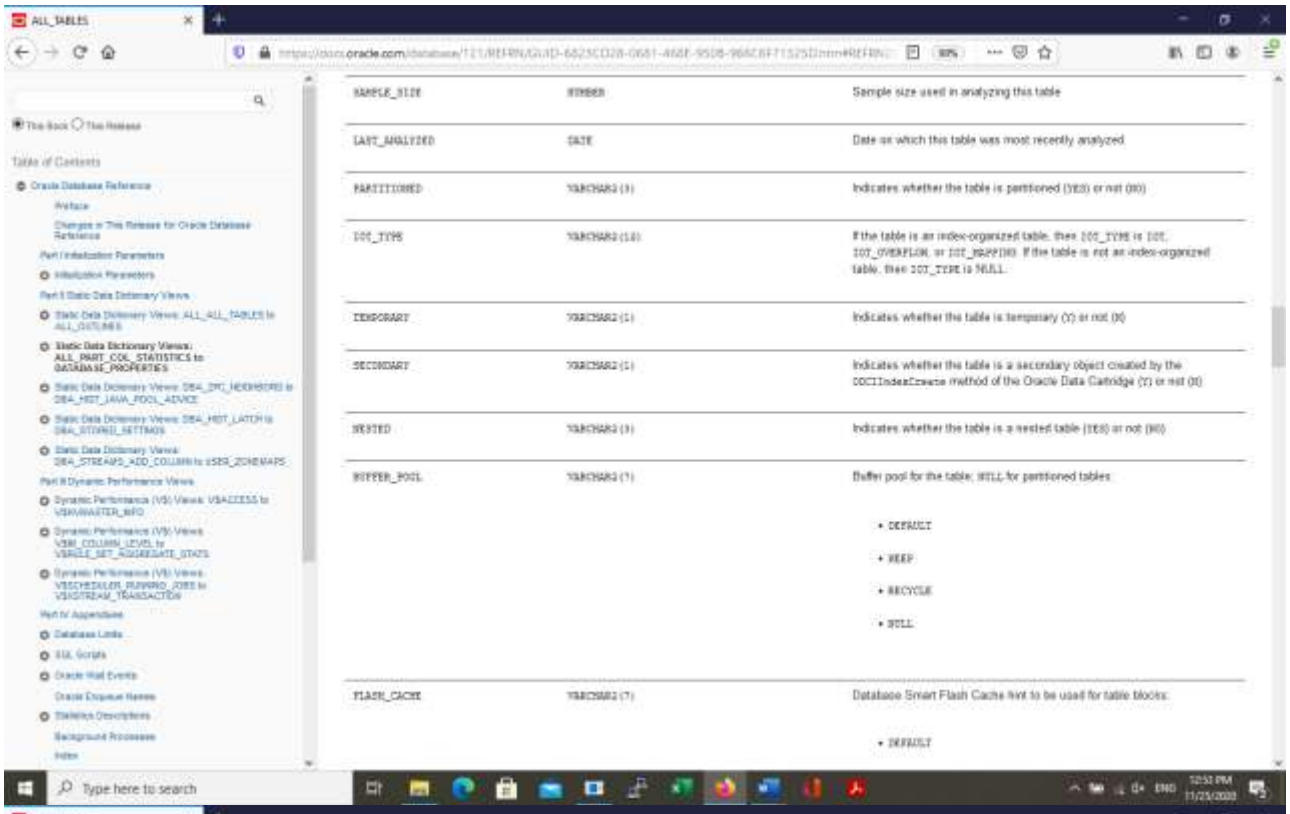




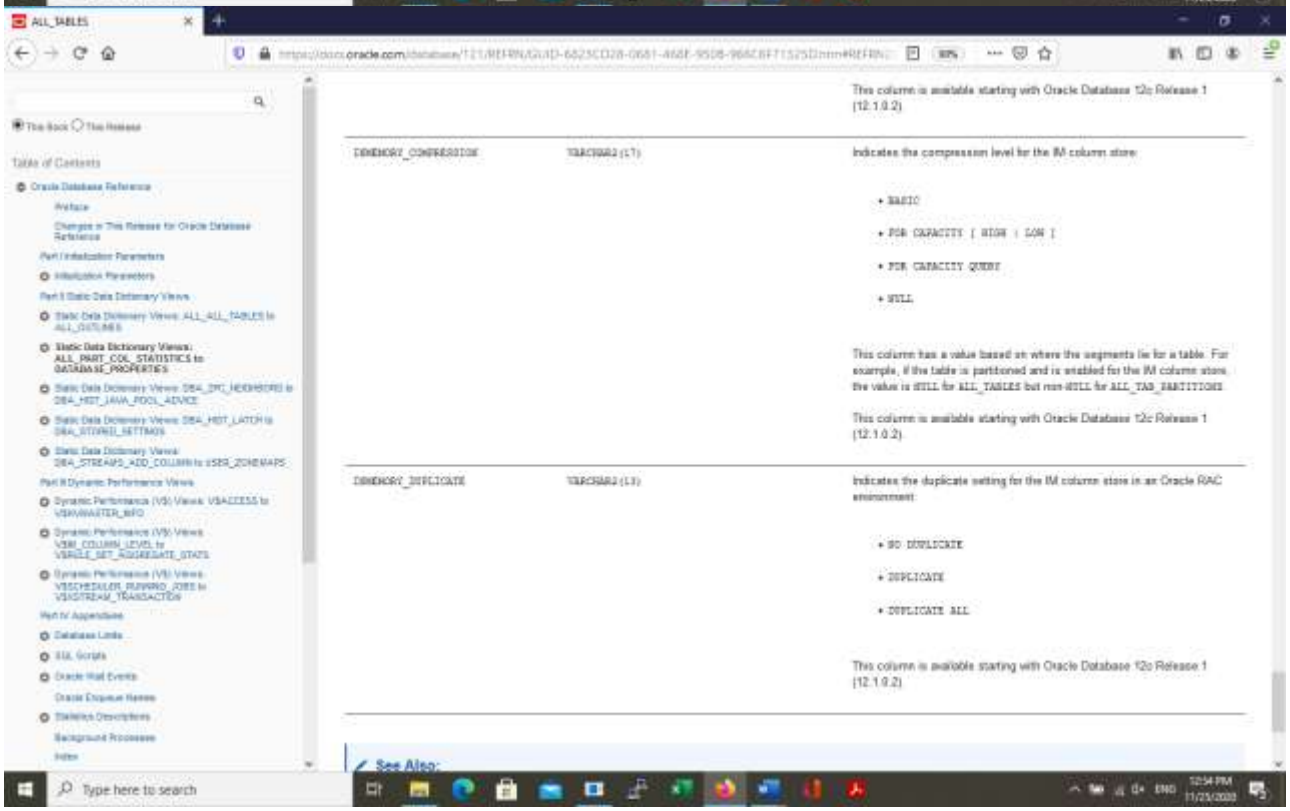
MAX_TRANS	NUMBER	Maximum number of transactions; NULL for partitioned tables
INITIAL_EXTENT	NUMBER	Size of the initial extent (in bytes); NULL for partitioned tables
NEXT_EXTENT	NUMBER	Size of secondary extents (in bytes); NULL for partitioned tables
MIN_EXTENTS	NUMBER	Minimum number of extents allowed in the segment; NULL for partitioned tables
MAX_EXTENTS	NUMBER	Maximum number of extents allowed in the segment; NULL for partitioned tables
PCT_INCREASE	NUMBER	Percentage increase in extent size; NULL for partitioned tables
FREELISTS	NUMBER	Number of process freelists allocated to the segment; NULL for partitioned tables
FREELIST_GROUPS	NUMBER	Number of freelist groups allocated to the segment; NULL for partitioned tables
LOGGING	VARCHAR2 (3)	Indicates whether or not changes to the table are logged; NULL for partitioned tables: • YES • NO
BACKED_UP	VARCHAR2 (1)	Indicates whether the table has been backed up since the last modification (Y) or not (N)



NUM_ROWS*	NUMBER	Number of rows in the table
BLOCKS*	NUMBER	Number of used data blocks in the table
EMPTY_BLOCKS	NUMBER	Number of empty (never used) data blocks in the table. This column is populated only if you collect statistics on the table using the DBMS_STATS package
AVG_SPACE*	NUMBER	Average amount of free space, in bytes, in a data block allocated to the table
CHAIN_CNT*	NUMBER	Number of rows in the table that are chained from one data block to another, or which have migrated to a new block, requiring a link to preserve the old ROWID
AVG_ROW_LEN*	NUMBER	Average length of a row in the table (in bytes)
AVG_SPACE_FREELIST_BLOCKS	NUMBER	Average freespace of all blocks on a freelist
NUM_FREELIST_BLOCKS	NUMBER	Number of blocks on the freelist
DEGREE	VARCHAR2 (10)	Number of threads per instance for scanning the table, or DEFAULT
INSTANCES	VARCHAR2 (10)	Number of instances across which the table is to be scanned, or DEFAULT
CACHE	VARCHAR2 (5)	Indicates whether the table is to be cached in the buffer cache (Y) or not (N)
TABLE_LOCK	VARCHAR2 (8)	Indicates whether table locking is enabled (ENABLED) or disabled (DISABLED)



SAMPLE_SIZE	NUMBER	Sample size used in analyzing this table
LAST_ANALYZED	DATE	Date on which this table was most recently analyzed
PARTITIONED	VARCHAR2 (3)	Indicates whether the table is partitioned (YES) or not (NO)
IOT_TYPE	VARCHAR2 (12)	If the table is an index-organized table, then IOT_TYPE is IOT, IOT_OVERFLOW, or IOT_MAPPING. If the table is not an index-organized table, then IOT_TYPE is NULL.
TEMPORARY	VARCHAR2 (1)	Indicates whether the table is temporary (Y) or not (N)
SECONDARY	VARCHAR2 (1)	Indicates whether the table is a secondary object created by the ODCIIndexCreate method of the Oracle Data Cartridge (Y) or not (N)
NESTED	VARCHAR2 (3)	Indicates whether the table is a nested table (YES) or not (NO)
BUFFER_POOL	VARCHAR2 (7)	Buffer pool for the table; NULL for partitioned tables • DEFAULT • KEEP • RECYCLE • NULL
FLASH_CACHE	VARCHAR2 (7)	Database Smart Flash Cache hint to be used for table blocks: • DEFAULT



		This column is available starting with Oracle Database 12c Release 1 (12.1.0.2)
INMEMORY_COMPRESSION	VARCHAR2 (17)	Indicates the compression level for the IM column store: • NOMEMCOMPRESS • FOR CAPACITY [HIGH | LOW] • FOR CAPACITY QUERY • NULL This column has a value based on where the segments lie for a table. For example, if the table is partitioned and is enabled for the IM column store, the value is NULL for ALL_TABLES but non-NULL for ALL_TAB_PARTITIONS This column is available starting with Oracle Database 12c Release 1 (12.1.0.2)
INMEMORY_DUPLICATE	VARCHAR2 (13)	Indicates the duplicate setting for the IM column store in an Oracle RAC environment: • NO DUPLICATE • DUPLICATE • DUPLICATE ALL This column is available starting with Oracle Database 12c Release 1 (12.1.0.2)

See Also:

```
SQL> select owner, tablespace_name, table_name  from dba_tables
        where owner = 'SCOTT';

OWNER                          TABLESPACE_NAME                TABLE_NAME
------------------------------ ------------------------------ ------------
SCOTT                                USERS                    CATALOG
SCOTT                                USERS                    DISCIPLINE
SCOTT                                USERS                    DEPT
SCOTT                                USERS                    EMP
SCOTT                                USERS                    BONUS
SCOTT                                USERS                    SALGRADE
```

13) Informatii despre tabelele create de userul curent :

```
SQL> desc user_tables

 Name                                      Null?                Type
 ----------------------------------------- --------             ------
 TABLE_NAME                                NOT NULL             VARCHAR2(30)
 TABLESPACE_NAME                                                VARCHAR2(30)
 CLUSTER_NAME                                                   VARCHAR2(30)
 IOT_NAME                                                       VARCHAR2(30)
 STATUS                                                         VARCHAR2(8)
 PCT_FREE                                                       NUMBER
 PCT_USED                                                       NUMBER
 INI_TRANS                                                      NUMBER
 MAX_TRANS                                                      NUMBER
 INITIAL_EXTENT                                                 NUMBER
 NEXT_EXTENT                                                    NUMBER
 MIN_EXTENTS                                                    NUMBER
 MAX_EXTENTS                                                    NUMBER
 PCT_INCREASE                                                   NUMBER
 FREELISTS                                                      NUMBER
 FREELIST_GROUPS                                                NUMBER
 LOGGING                                                        VARCHAR2(3)
 BACKED_UP                                                      VARCHAR2(1)
 NUM_ROWS                                                       NUMBER
 BLOCKS                                                         NUMBER
 EMPTY_BLOCKS                                                   NUMBER
 AVG_SPACE                                                      NUMBER
 CHAIN_CNT                                                      NUMBER
 AVG_ROW_LEN                                                    NUMBER
 AVG_SPACE_FREELIST_BLOCKS                                      NUMBER
 NUM_FREELIST_BLOCKS                                            NUMBER
 DEGREE                                                         VARCHAR2(10)
 INSTANCES                                                      VARCHAR2(10)
```

CACHE	VARCHAR2(5)
TABLE_LOCK	VARCHAR2(8)
SAMPLE_SIZE	NUMBER
LAST_ANALYZED	DATE
PARTITIONED	VARCHAR2(3)
IOT_TYPE	VARCHAR2(12)
TEMPORARY	VARCHAR2(1)
SECONDARY	VARCHAR2(1)
NESTED	VARCHAR2(3)
BUFFER_POOL	VARCHAR2(7)
ROW_MOVEMENT	VARCHAR2(8)
GLOBAL_STATS	VARCHAR2(3)
USER_STATS	VARCHAR2(3)
DURATION	VARCHAR2(15)
SKIP_CORRUPT	VARCHAR2(8)
MONITORING	VARCHAR2(3)
CLUSTER_OWNER	VARCHAR2(30)
DEPENDENCIES	VARCHAR2(8)
COMPRESSION	VARCHAR2(8)
DROPPED	VARCHAR2(3)
READ_ONLY	VARCHAR2(3)
SEGMENT_CREATED	VARCHAR2(3)
RESULT_CACHE	VARCHAR2(7)
CLUSTERING	VARCHAR2(3)
ACTIVITY_TRACKING	VARCHAR2(23)
DML_TIMESTAMP	VARCHAR2(25)
HAS_IDENTITY	VARCHAR2(3)
CONTAINER_DATA	VARCHAR2(3)
INMEMORY	VARCHAR2(8)
INMEMORY_PRIORITY	VARCHAR2(8)
INMEMORY_DISTRIBUTE	VARCHAR2(15)
INMEMORY_COMPRESSION	VARCHAR2(17)
INMEMORY_DUPLICATE	VARCHAR2(13)

SQL> select instances, tablespace_name, table_name from user_tables;

INSTANCES	TABLESPACE_NAME	TABLE_NAME
1	BD_DATA	ANGAJATI
1		ANGAJATI_DEP
1	BD_DATA	DEPT
1	BD_DATA	EMP
1		EMP_TEMP
1	BD_DATA	BONUS
1	BD_DATA	SALGRADE
1	BD_DATA	STUDENTI

1 BD_DATA CATALOG

14) Informatii despre tabelele create de userul curent :

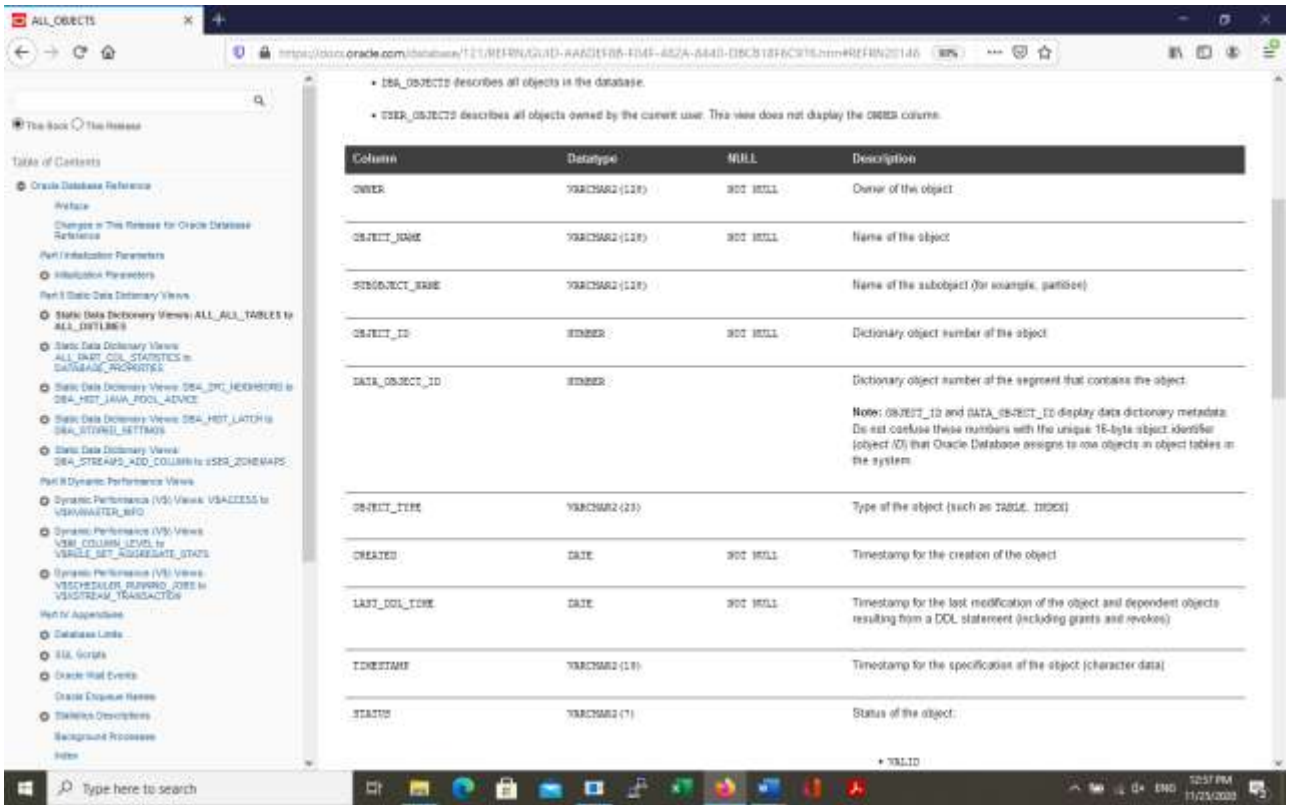
```
SQL> select tablespace_name, table_name, partition_name, high_value from user_tab_partitions
        where table_name = 'ANGAJATI_DEP'
         order by partition_name;
```

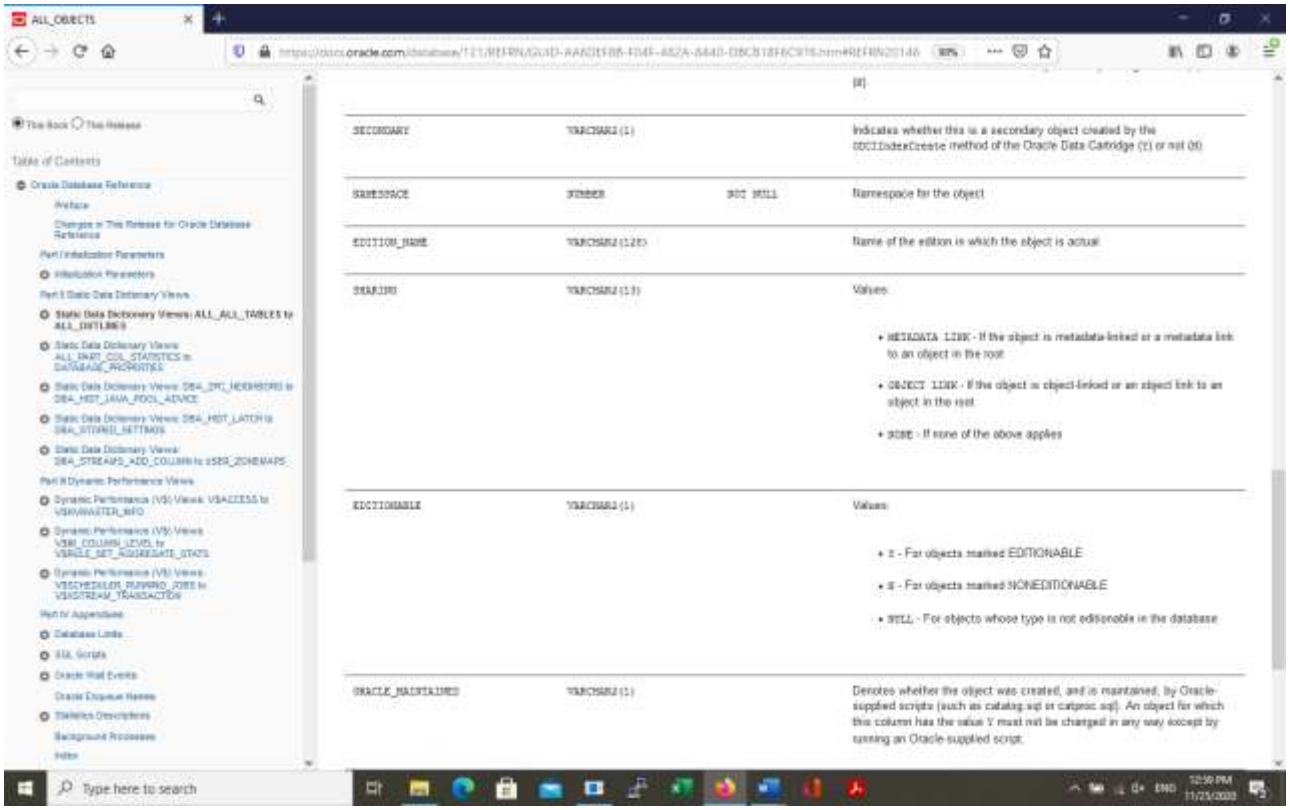
TABLESPACE_NAME	TABLE_NAME	PARTITION_NAME	HIGH_VALUE
UBD_DATA	ANGAJATI_DEP	DEP10	10
UBD_DATA	ANGAJATI_DEP	DEP20	20
UBD_DATA	ANGAJATI_DEP	DEP30	30

15) Informatii despre obiectele din baza de date:

```
SQL> desc dba_objects
```

Nume	Nul?	Type
OWNER		VARCHAR2(30)
OBJECT_NAME		VARCHAR2(128)
SUBOBJECT_NAME		VARCHAR2(30)
OBJECT_ID		NUMBER
DATA_OBJECT_ID		NUMBER
OBJECT_TYPE		VARCHAR2(18)
CREATED		DATE
LAST_DDL_TIME		DATE
TIMESTAMP		VARCHAR2(19)
STATUS		VARCHAR2(7)
TEMPORARY		VARCHAR2(1)
GENERATED		VARCHAR2(1)
SECONDARY		VARCHAR2(1)
NAMESPACE		NUMBER
EDITION_NAME		VARCHAR2(128)
SHARING		VARCHAR2(13)
EDITIONABLE		VARCHAR2(1)
ORACLE_MAINTAINED		VARCHAR2(1)
SECONDARY		VARCHAR2(1)


ALL_OBJECTS
• DBA_OBJECTS describes all objects in the database.
• USER_OBJECTS describes all objects owned by the current user. This view does not display the OWNER column.
Column | Datatype | NULL | Description
OWNER | VARCHAR2(128) | NOT NULL | Owner of the object
OBJECT_NAME | VARCHAR2(128) | NOT NULL | Name of the object
SUBOBJECT_NAME | VARCHAR2(128) | | Name of the subobject (for example, partition)
OBJECT_ID | NUMBER | NOT NULL | Dictionary object number of the object
DATA_OBJECT_ID | NUMBER | | Dictionary object number of the segment that contains the object
Note: OBJECT_ID and DATA_OBJECT_ID display data dictionary metadata. Do not confuse these numbers with the unique 16-byte object identifier (object ID) that Oracle Database assigns to row objects in object tables or the system
OBJECT_TYPE | VARCHAR2(23) | | Type of the object (such as TABLE, INDEX)
CREATED | DATE | NOT NULL | Timestamp for the creation of the object
LAST_DDL_TIME | DATE | NOT NULL | Timestamp for the last modification of the object and dependent objects resulting from a DDL statement (including grants and revokes)
TIMESTAMP | VARCHAR2(19) | | Timestamp for the specification of the object (character data)
STATUS | VARCHAR2(7) | | Status of the object:
• VALID
Type here to search


ALL_OBJECTS
SECONDARY | VARCHAR2(1) | | Indicates whether this is a secondary object created by the ODCIIndexCreate method of the Oracle Data Cartridge (Y) or not (N)
NAMESPACE | NUMBER | NOT NULL | Namespace for the object
EDITION_NAME | VARCHAR2(128) | | Name of the edition in which the object is actual
SHARING | VARCHAR2(13) | | Values:
• METADATA LINK - If the object is metadata-linked or a metadata link to an object in the root
• OBJECT LINK - If the object is object-linked or an object link to an object in the root
• NONE - If none of the above applies
EDITIONABLE | VARCHAR2(1) | | Values:
• Y - For objects marked EDITIONABLE
• N - For objects marked NONEDITIONABLE
• NULL - For objects whose type is not editionable in the database
ORACLE_MAINTAINED | VARCHAR2(1) | | Denotes whether the object was created, and is maintained, by Oracle-supplied scripts (such as catalog.sql or catproc.sql). An object for which this column has the value Y must not be changed in any way except by running an Oracle-supplied script.
Type here to search

```
SQL> select object_name, created
        from dba_objects
        where object_name like 'EMP%' and owner ='SCOTT';

OBJECT_NAME          CREATED
-------------------- ----------
EMP                  07-MAY-2019
EMP_COPY             01-NOV-2019
```

16) Informatii despre obiectele create de userul current:

```
SQL> desc user_objects

Name                                                      Null?    Type
 ----------------------------------------------------------------- -------- ------
 OBJECT_NAME                                                       VARCHAR2(128)
 SUBOBJECT_NAME                                                    VARCHAR2(30)
 OBJECT_ID                                                         NUMBER
 DATA_OBJECT_ID                                                    NUMBER
 OBJECT_TYPE                                                       VARCHAR2(19)
 CREATED                                                           DATE
 LAST_DDL_TIME                                                     DATE
 TIMESTAMP                                                         VARCHAR2(19)
 STATUS                                                            VARCHAR2(7)
 TEMPORARY                                                         VARCHAR2(1)
 GENERATED                                                         VARCHAR2(1)
 SECONDARY                                                         VARCHAR2(1)
 NAMESPACE                                                         NUMBER
 EDITION_NAME                                                      VARCHAR2(128)
 SHARING                                                           VARCHAR2(13)
 EDITIONABLE                                                       VARCHAR2(1)
 ORACLE_MAINTAINED                                                 VARCHAR2(1)
 SECONDARY                                                         VARCHAR2(1)
```

```
SQL>  select object_name, object_type,created, status from user_objects;

OBJECT_NAME        OBJECT_TYPE          CREATED          STATUS
-------------------- -------------------  ---------        -------
PK_DEPT              INDEX                17-APR-07        VALID
DEPT                 TABLE                17-APR-07        VALID
EMP                  TABLE                17-APR-07        VALID
PK_EMP               INDEX                17-APR-07        VALID
BONUS                TABLE                17-APR-07        VALID
SALGRADE             TABLE                17-APR-07        VALID
V_STOCURI2           VIEW                 13-MAR-15        INVALID
CG$DISCIPLINE        PACKAGE              04-MAR-14        VALID
```

CG$DISCIPLINE	PACKAGE BODY	04-MAR-14	VALID
CG$BIS_STUDENTI	TRIGGER	04-MAR-14	VALID

17) Stergerea datelor dintr-o tabela, cu posibilitate de rollback:

```
SQL> delete from  emp_copy where deptno=10;
```

18) Trunchierea unei tabele (stergerea datelor dintr-o tabela, cu eliberarea spatiului ocupat de tabela si fara posibilitate de rollback):

```
SQL> truncate table emp_copy;
```

19) Stergerea unei tabele din dictionar, cu stergerea tuturor referintelor pe tabela:

```
SQL> drop table emp_copy cascade constraints;
```

20) Stergerea unei linii dintr-o tabela care contine linii duplicat folosind ROWID:

```
SQL> create table emp_del as select * from emp;
Table created.

SQL> insert into emp_del  select * from emp;
14 rows created.

SQL> select  rowid, empno, ename from emp_del order by empno;
```

ROWID	EMPNO	ENAME
AAAQwSAAEAAAAKUAAA	7369	SMITH
AAAQwSAAEAAAAKWAAA	7369	SMITH
AAAQwSAAEAAAAKWAAB	7499	ALLEN
AAAQwSAAEAAAAKUAAB	7499	ALLEN
AAAQwSAAEAAAAKUAAC	7521	WARD
AAAQwSAAEAAAAKWAAC	7521	WARD
AAAQwSAAEAAAAKWAAD	7566	JONES
AAAQwSAAEAAAAKUAAD	7566	JONES
AAAQwSAAEAAAAKUAAE	7654	MARTIN
AAAQwSAAEAAAAKWAAE	7654	MARTIN

……..

```
SQL> delete from emp_del where rowid='AAAQwSAAEAAAAKUAAB'; (ALLEN)
1 row deleted.
```

SQL> select rowid, empno, ename from emp_del order by empno;

ROWID	EMPNO	ENAME
AAAQwSAAEAAAAKUAAA	7369	SMITH
AAAQwSAAEAAAAKWAAA	7369	SMITH
AAAQwSAAEAAAAKWAAB	7499	ALLEN
AAAQwSAAEAAAAKUAAC	7521	WARD
AAAQwSAAEAAAAKWAAC	7521	WARD
AAAQwSAAEAAAAKWAAD	7566	JONES
AAAQwSAAEAAAAKUAAD	7566	JONES
AAAQwSAAEAAAAKUAAE	7654	MARTIN
AAAQwSAAEAAAAKWAAE	7654	MARTIN

21) Stergerea unei linii dintr-o tabela care contine duplicate folosind ROWNUM:

SQL> select rownum, empno, ename from emp_del order by empno;

ROWNUM	EMPNO	ENAME
1	7369	SMITH
15	7369	SMITH
16	7499	ALLEN
2	7499	ALLEN
3	7521	WARD
17	7521	WARD
18	7566	JONES
4	7566	JONES

……

SQL> delete from emp_del where rownum=16;
0 rows deleted.

SQL> delete from emp_del where rownum>16 and rownum<18;
0 rows deleted.

SQL> delete from emp_del where rownum<15;
14 rows deleted.

SQL> select rownum, empno, ename from emp_del order by empno;

ROWNUM	EMPNO	ENAME
1	7369	SMITH
2	7499	ALLEN
3	7521	WARD
4	7566	JONES
5	7654	MARTIN
6	7698	BLAKE
7	7782	CLARK
8	7788	SCOTT
9	7839	KING
10	7844	TURNER
11	7876	ADAMS
12	7900	JAMES
13	7902	FORD
14	7934	MILLER

14 rows selected.

SQL> create table emp_copy2 as select * from emp; -- cream o noua copie
Table created.

SQL> insert into emp_copy2 select * from emp; -- duplicam liniile in tabela emp_copy2
14 rows created.

SQL> select rownum, empno, ename from emp_copy2 order by empno;

ROWNUM	EMPNO	ENAME
1	7369	SMITH
15	7369	SMITH
16	7499	ALLEN
2	7499	ALLEN
3	7521	WARD
17	7521	WARD
18	7566	JONES
4	7566	JONES
5	7654	MARTIN
19	7654	MARTIN
6	7698	BLAKE
20	7698	BLAKE
7	7782	CLARK
21	7782	CLARK
8	7788	SCOTT
22	7788	SCOTT
9	7839	KING
23	7839	KING

10	7844	TURNER
24	7844	TURNER
11	7876	ADAMS
25	7876	ADAMS
12	7900	JAMES
26	7900	JAMES
13	7902	FORD
27	7902	FORD
28	7934	MILLER
14	7934	MILLER

28 rows selected.

SQL> delete from emp_copy2 where rownum=(select min(rownum) from emp_copy2 where empno=7566); (JONES)
1 row deleted.

SQL> select rownum, empno, ename from emp_copy2 order by empno;

ROWNUM	EMPNO	ENAME
14	7369	SMITH
1	7499	ALLEN
15	7499	ALLEN
2	7521	WARD
16	7521	WARD
17	7566	JONES
3	7566	JONES
4	7654	MARTIN
18	7654	MARTIN
5	7698	BLAKE
19	7698	BLAKE
6	7782	CLARK
20	7782	CLARK
7	7788	SCOTT
21	7788	SCOTT
8	7839	KING
22	7839	KING
9	7844	TURNER
23	7844	TURNER
10	7876	ADAMS
24	7876	ADAMS
11	7900	JAMES
25	7900	JAMES
12	7902	FORD
26	7902	FORD

```
13          7934    MILLER
27          7934    MILLER

27 rows selected.
```

Obs. In tabela emp_copy2 a fost stearsa linia cu rownum=1 (SMITH) in loc de linia cu rownum=4 (JONES) !! Explicatia este ca subcererea lucreaza astfel:

```
SQL> select rownum from emp_copy2 where empno=7566;

   ROWNUM
----------
         1
         2

SQL>  select min(rownum) from emp_copy2 where empno=7566;

MIN(ROWNUM)
-----------
          1
```

Concluzii:

a) Pentru stergerea manuala a liniilor se poate folosi ROWID dar Oracle nu recomanda aceasta metoda.

b) Se poate folosi stergerea in bloc a liniilor duplicat folosind ROWNUM, in anumite cazuri particulare, dar cu mare atentie. Nici aceasta metoda nu este recomandata de catre Oracle.

c) Pentru stergerea liniilor duplicat se recomanda utilizarea executiilor ciclice din PL/SQL (LOOP, FOR, WHILE, cursoare, etc).

22) Stergerea **unei singure linii duplicat dintr-o tabela care contine duplicate,** folosind subcereri:

```
SQL> select * from emp_copy2 A where rowid
        IN (select min(rowid) from emp_copy2 B where ename='SMITH');

     EMPNO       ENAME      JOB           MGR   HIREDATE        SAL      COMM     DEPTNO
----------       ---------- ---------     ---------- ---------  ----------  ----------  ----------
      7369       SMITH      CLERK         7902   17-DEC-80        800                  20

SQL> delete from emp_copy2 A where rowid IN
        (select min(rowid) from emp_copy2 B where ename='WARD');
 1 row deleted.
```

SQL> select rownum, empno, ename from emp_copy2 order by empno;

ROWNUM	EMPNO	ENAME
1	7369	SMITH
14	7369	SMITH
2	7499	ALLEN
15	7499	ALLEN
16	7521	WARD
3	7566	JONES
17	7566	JONES
4	7654	MARTIN
18	7654	MARTIN

23. Stergerea **liniilor duplicat dintr-o tabela care contine cate un duplicat la fiecare linie**, folosind subcereri:

```
SQL> select * from emp_copy2 A where
        rowid IN (select min(rowid) from emp_copy2 B where A.empno=B.empno) order by ename;

SQL>  delete  from emp_copy2 A where
         rowid IN (select min(rowid) from emp_copy2 B where A.empno=B.empno);
```

14 rows deleted.

24. Stergerea **tuturor liniilor duplicat dintr-o tabela care contine mai multe duplicate**, folosind subcereri:

```
SQL> select * from emp_copy2 A where
        rowid > (select min(rowid) from emp_copy2 B where A.empno=B.empno) order by ename;

SQL> delete  from emp_copy2 A where
        rowid >(select min(rowid) from emp_copy2 B where A.empno=B.empno);
```

28 rows deleted.

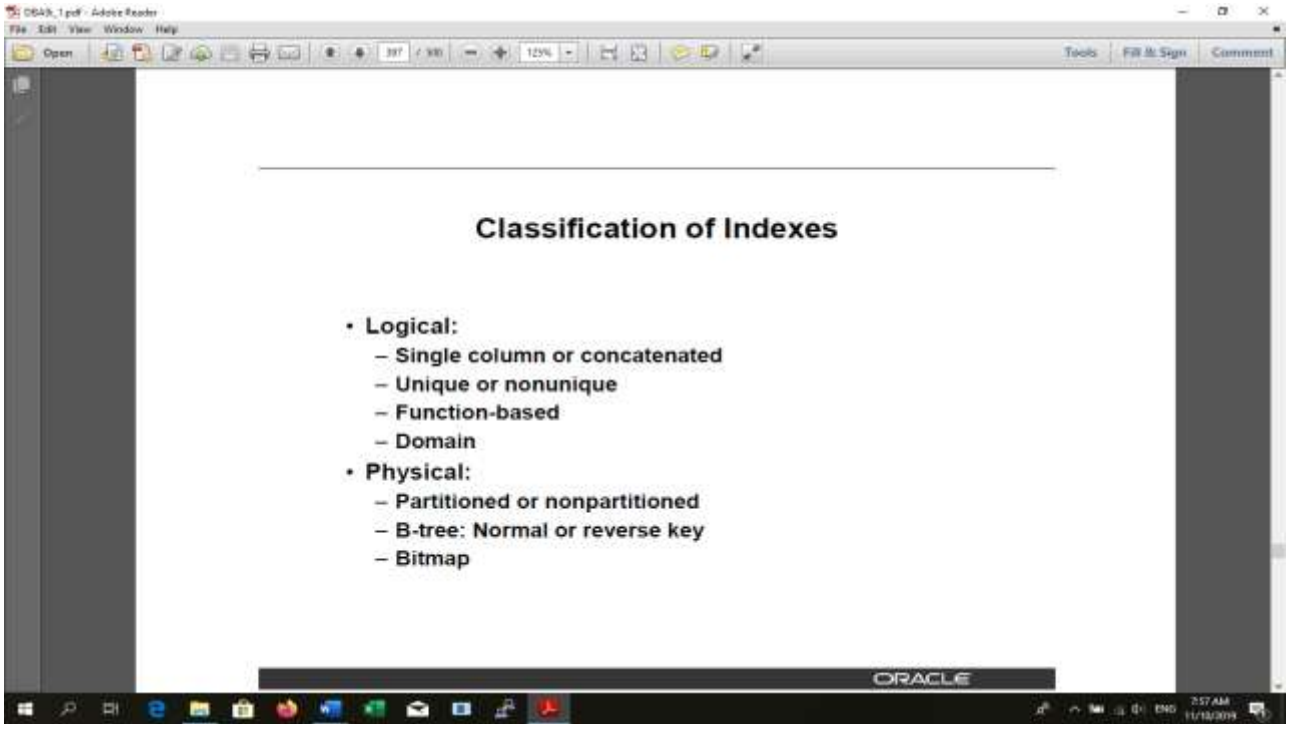
SQL> select rownum, empno, ename from emp_copy2 order by empno;

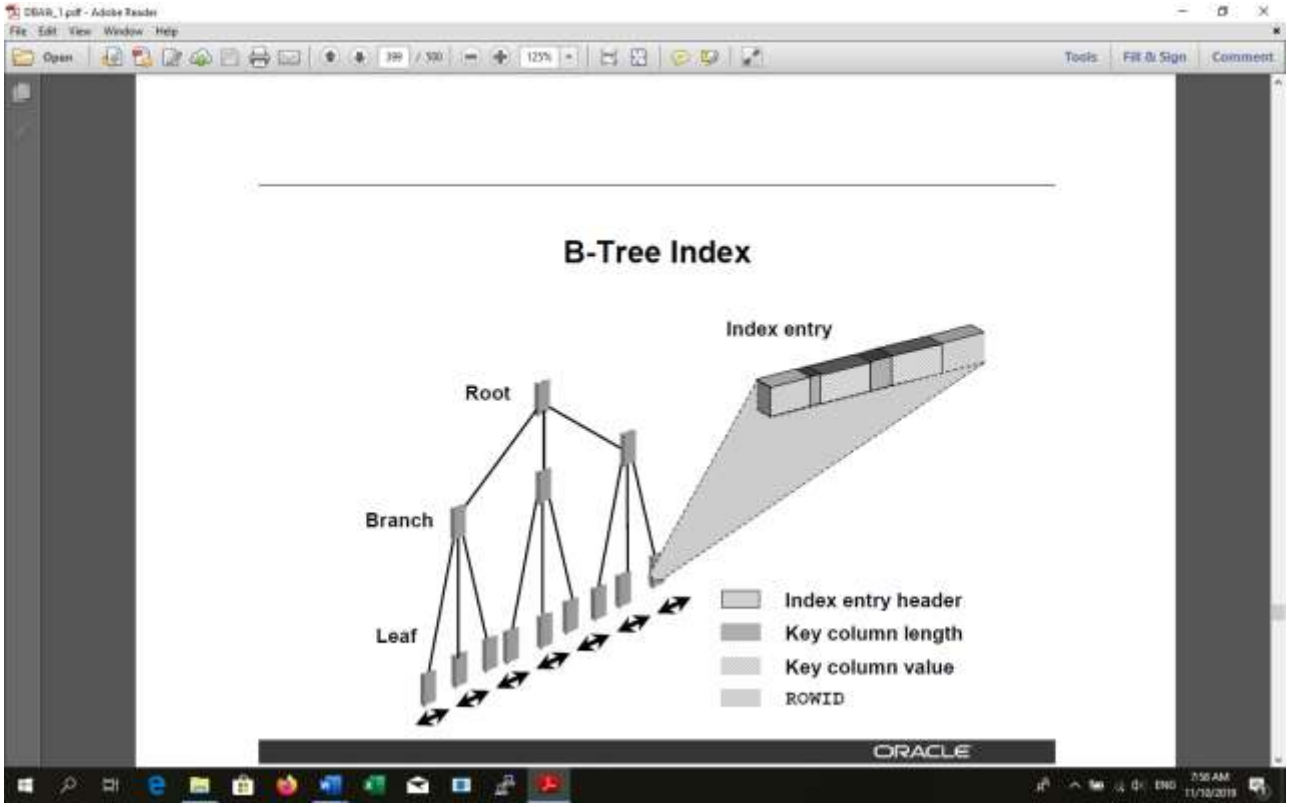
ROWNUM	EMPNO	ENAME
1	7369	SMITH
2	7499	ALLEN
3	7521	WARD
4	7566	JONES
5	7654	MARTIN

6	7698	BLAKE
7	7782	CLARK
8	7788	SCOTT
9	7839	KING
10	7844	TURNER
11	7876	ADAMS
12	7900	JAMES
13	7902	FORD
14	7934	MILLER

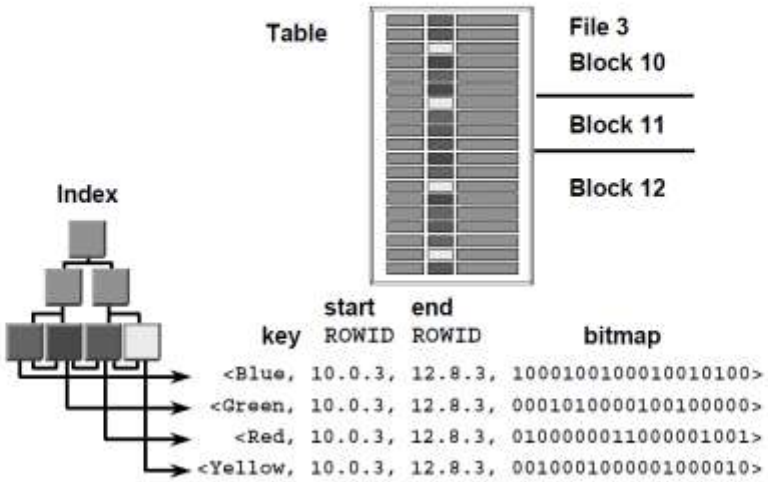
14 rows selected.

B. Indecsi


Classification of Indexes
• Logical:
– Single column or concatenated
– Unique or nonunique
– Function-based
– Domain
• Physical:
– Partitioned or nonpartitioned
– B-tree: Normal or reverse key
– Bitmap
ORACLE

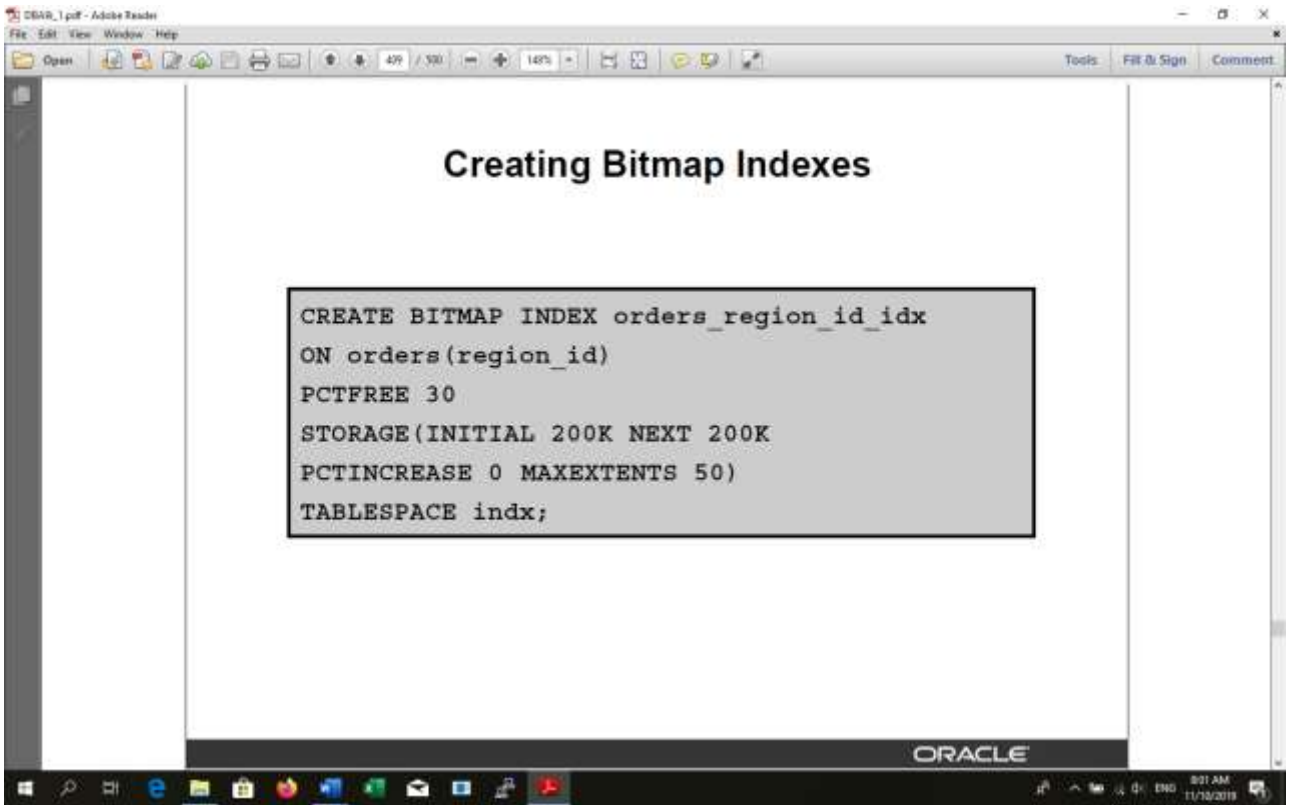

B-Tree Index
Index entry
Root
Branch
Leaf
Index entry header
Key column length
Key column value
ROWID
ORACLE

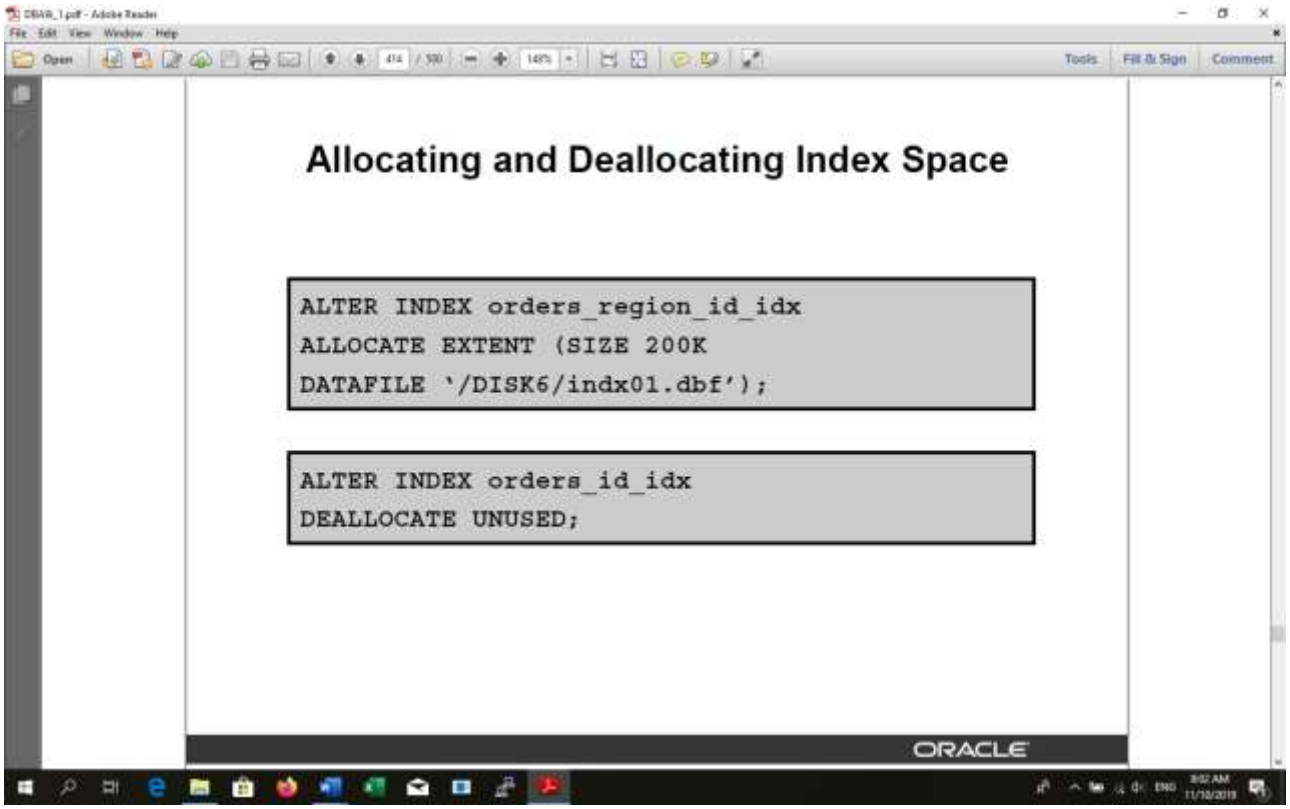
Bitmap Indexes



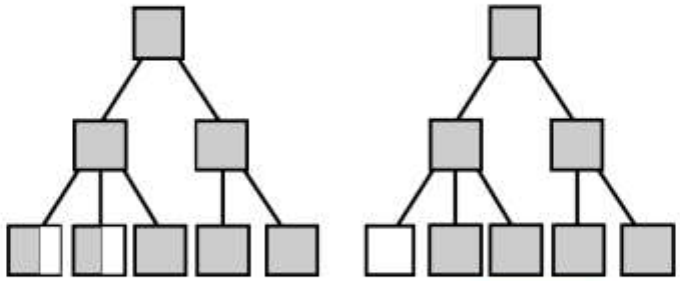
Creating B-Tree Indexes

```
CREATE INDEX hr.employees_last_name_idx
ON hr.employees(last_name)
PCTFREE 30
STORAGE(INITIAL 200K NEXT 200K
PCTINCREASE 0 MAXEXTENTS 50)
TABLESPACE indx;
```


Creating Bitmap Indexes
CREATE BITMAP INDEX orders_region_id_idx
ON orders(region_id)
PCTFREE 30
STORAGE(INITIAL 200K NEXT 200K
PCTINCREASE 0 MAXEXTENTS 50)
TABLESPACE indx;
ORACLE


Allocating and Deallocating Index Space
ALTER INDEX orders_region_id_idx
ALLOCATE EXTENT (SIZE 200K
DATAFILE '/DISK6/indx01.dbf');
ALTER INDEX orders_id_idx
DEALLOCATE UNUSED;
ORACLE

Coalescing Indexes



Before coalescing

After coalescing

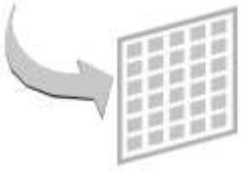
```
ALTER INDEX orders_id_idx COALESCE;
```

ORACLE

Checking Index Validity



```
ANALYZE INDEX orders_region_id_idx
VALIDATE STRUCTURE;
```

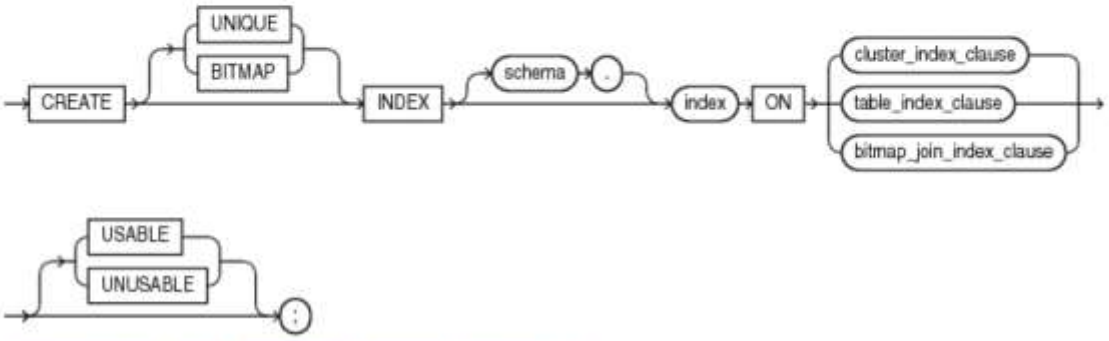


INDEX_STATS

ORACLE

Syntax

create_index::=



Description of the illustration "create_index.gif"



1) Crearea unui index de tip B-Tree:

```
SQL>  create index scott.emp_name_idx
          on scott.emp(ename)
          pctfree 30
          storage(initial 200k next 200k
          pctincrease 0 maxextents 50)
          tablespace bd_data;
Index created.
```

2) Crearea unui index de tip BITMAP:

```
SQL>  create  bitmap index scott.dept_name_idx
          on scott.dept(dname)
          pctfree 30
          storage(initial 200k next 200k
          pctincrease 0 maxextents 50)
          tablespace   bd_data;
Index created.
```

3) Alocarea unei extensii pentru un index de tip B-Tree:

```
SQL> alter tablespace users
          add datafile 'e:/temp/index01.dbf'  size 2M;
Tablespace altered.

SQL> alter index emp_name_idx
          allocate extent (size 200k datafile ’e:/temp/index01.dbf’)
Index altered.

SQL> alter tablespace users drop datafile 'e:/temp/index01.dbf';
ERROR at line 1:
ORA-03262: the file is non-empty
```

Obs. Dupa ce se aloca o extensie unui index fisierul respectiv nu se poate sterge. Trebuie sters mai intai indexul.

4) Eliberarea spatiului neutilizat pentru un index de tip B-Tree:

```
SQL> alter index emp_name_idx
         deallocate unused;
Index altered.
```

5) Mutarea unui index in alt tablespace:

```
SQL> alter index emp_name_idx rebuild tablespace  users;
Index altered.
```

6) Defragmentarea blocurilor unui index:

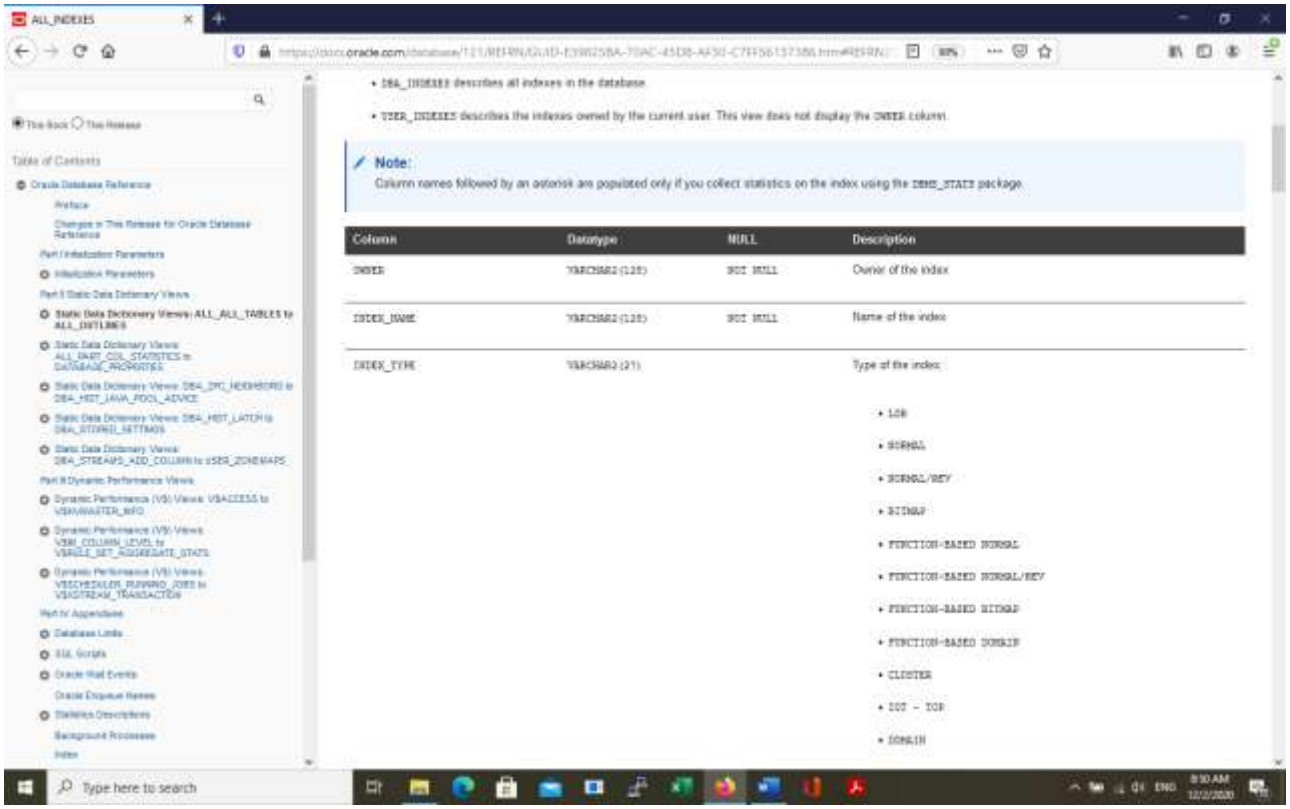
SQL> alter index emp_name_idx coalesce;
Index altered.

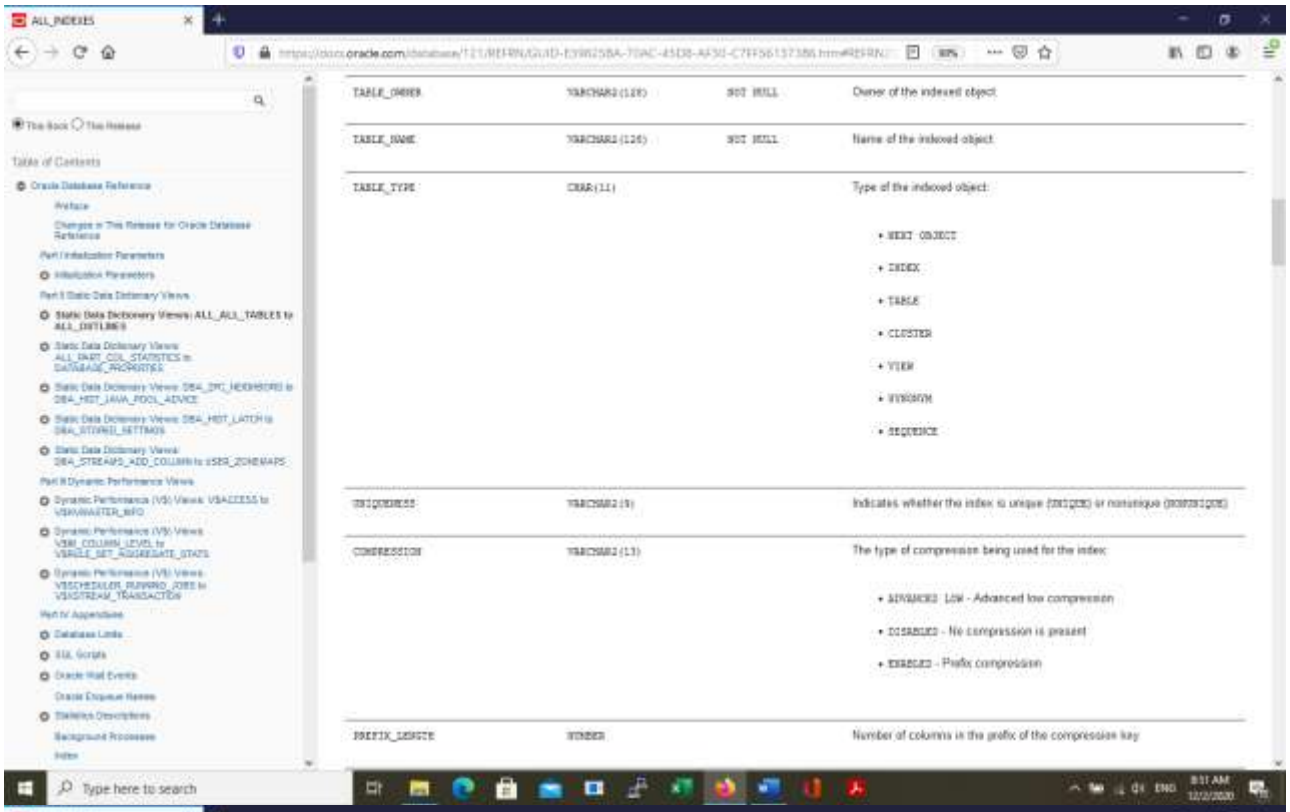
7) Informatii din dictionar despre indecsi:

SQL> desc dba_indexes

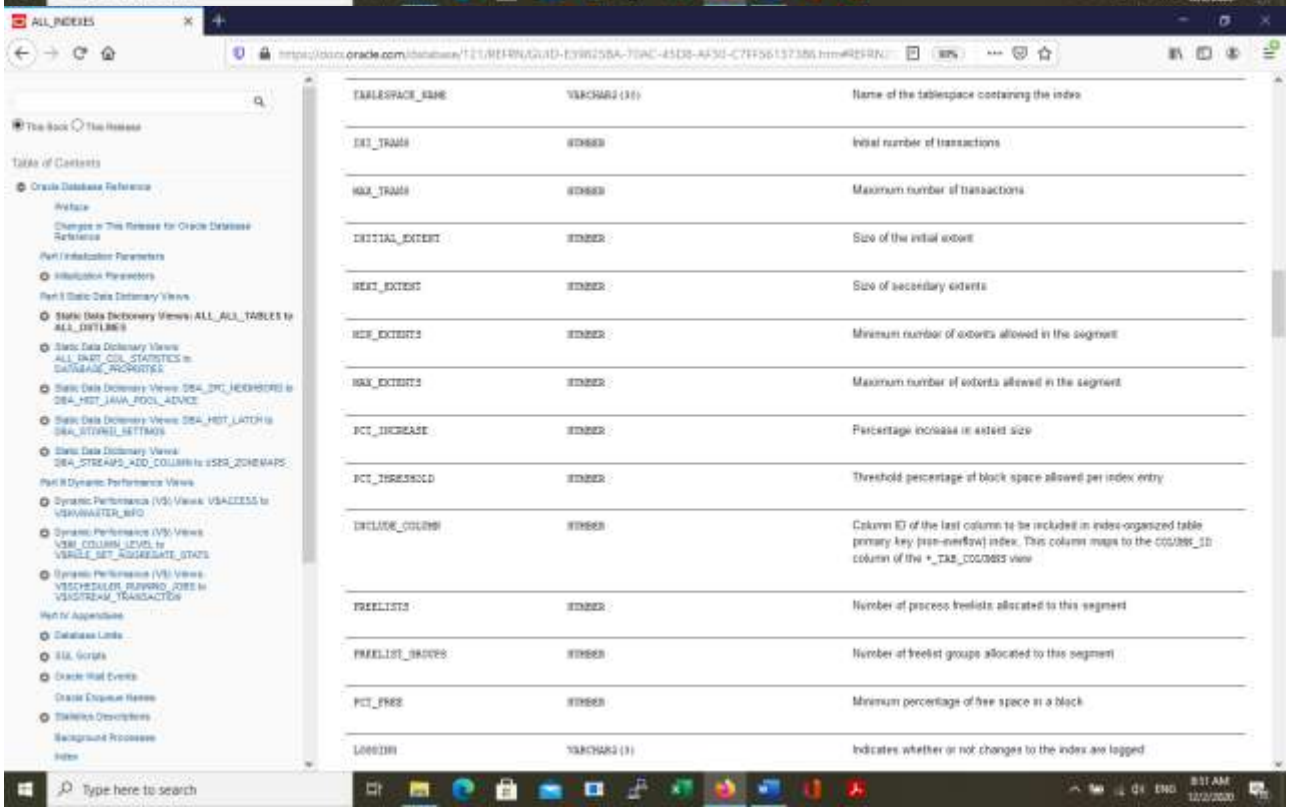
Name	Null?	Type
OWNER	NOT NULL	VARCHAR2(30)
INDEX_NAME	NOT NULL	VARCHAR2(30)
INDEX_TYPE		VARCHAR2(27)
TABLE_OWNER	NOT NULL	VARCHAR2(30)
TABLE_NAME	NOT NULL	VARCHAR2(30)
TABLE_TYPE		VARCHAR2(11)
UNIQUENESS		VARCHAR2(9)
COMPRESSION		VARCHAR2(8)
PREFIX_LENGTH		NUMBER
TABLESPACE_NAME		VARCHAR2(30)
INI_TRANS		NUMBER
MAX_TRANS		NUMBER
INITIAL_EXTENT		NUMBER
NEXT_EXTENT		NUMBER
MIN_EXTENTS		NUMBER
MAX_EXTENTS		NUMBER
PCT_INCREASE		NUMBER
PCT_THRESHOLD		NUMBER
INCLUDE_COLUMN		NUMBER
FREELISTS		NUMBER
FREELIST_GROUPS		NUMBER
PCT_FREE		NUMBER
LOGGING		VARCHAR2(3)
BLEVEL		NUMBER
LEAF_BLOCKS		NUMBER
DISTINCT_KEYS		NUMBER
AVG_LEAF_BLOCKS_PER_KEY		NUMBER
AVG_DATA_BLOCKS_PER_KEY		NUMBER
CLUSTERING_FACTOR		NUMBER
STATUS		VARCHAR2(8)
NUM_ROWS		NUMBER
SAMPLE_SIZE		NUMBER
LAST_ANALYZED		DATE
DEGREE		VARCHAR2(40)
INSTANCES		VARCHAR2(40)

PARTITIONED	VARCHAR2(3)
TEMPORARY	VARCHAR2(1)
GENERATED	VARCHAR2(1)
SECONDARY	VARCHAR2(1)
BUFFER_POOL	VARCHAR2(7)
USER_STATS	VARCHAR2(3)
DURATION	VARCHAR2(15)
PCT_DIRECT_ACCESS	NUMBER
ITYP_OWNER	VARCHAR2(30)
ITYP_NAME	VARCHAR2(30)
PARAMETERS	VARCHAR2(1000)
GLOBAL_STATS	VARCHAR2(3)
DOMIDX_STATUS	VARCHAR2(12)
DOMIDX_OPSTATUS	VARCHAR2(6)
FUNCIDX_STATUS	VARCHAR2(8)
JOIN_INDEX	VARCHAR2(3)

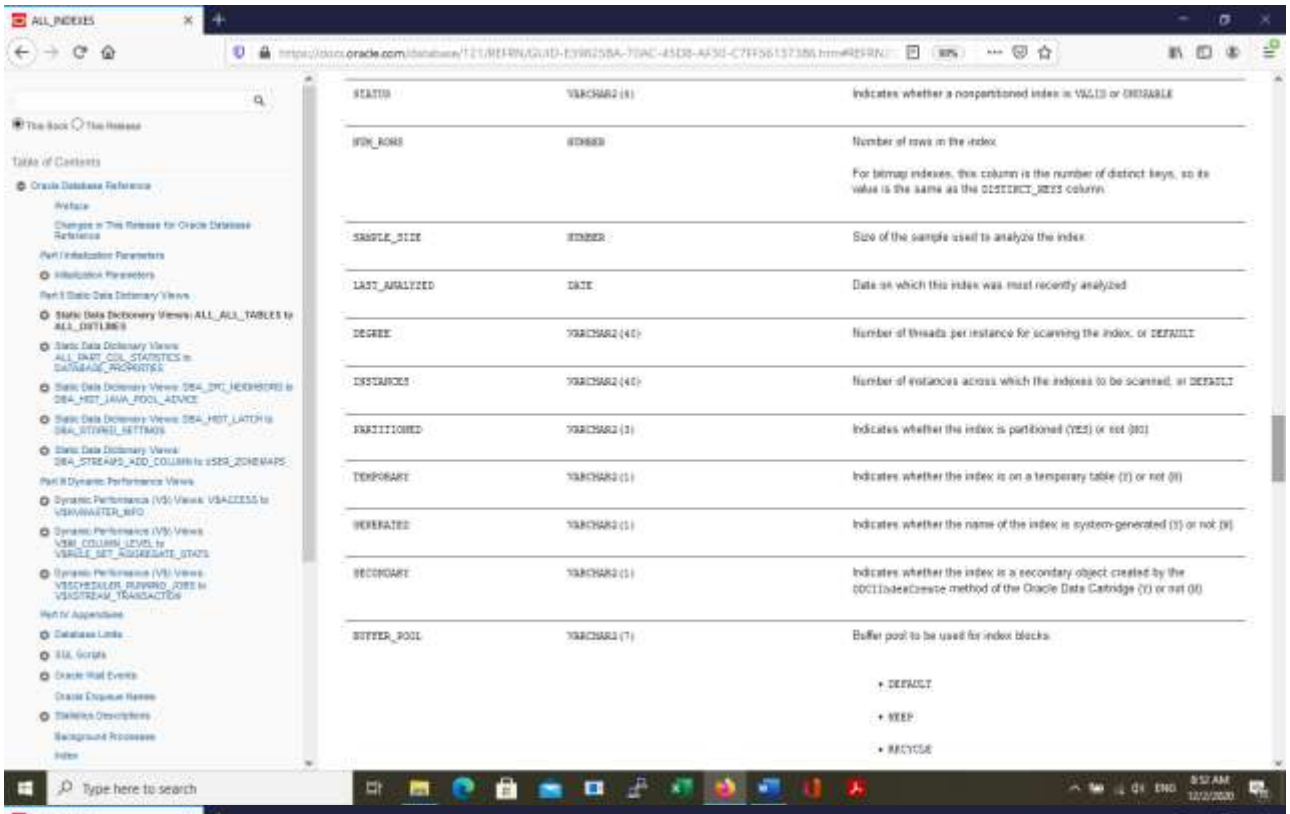




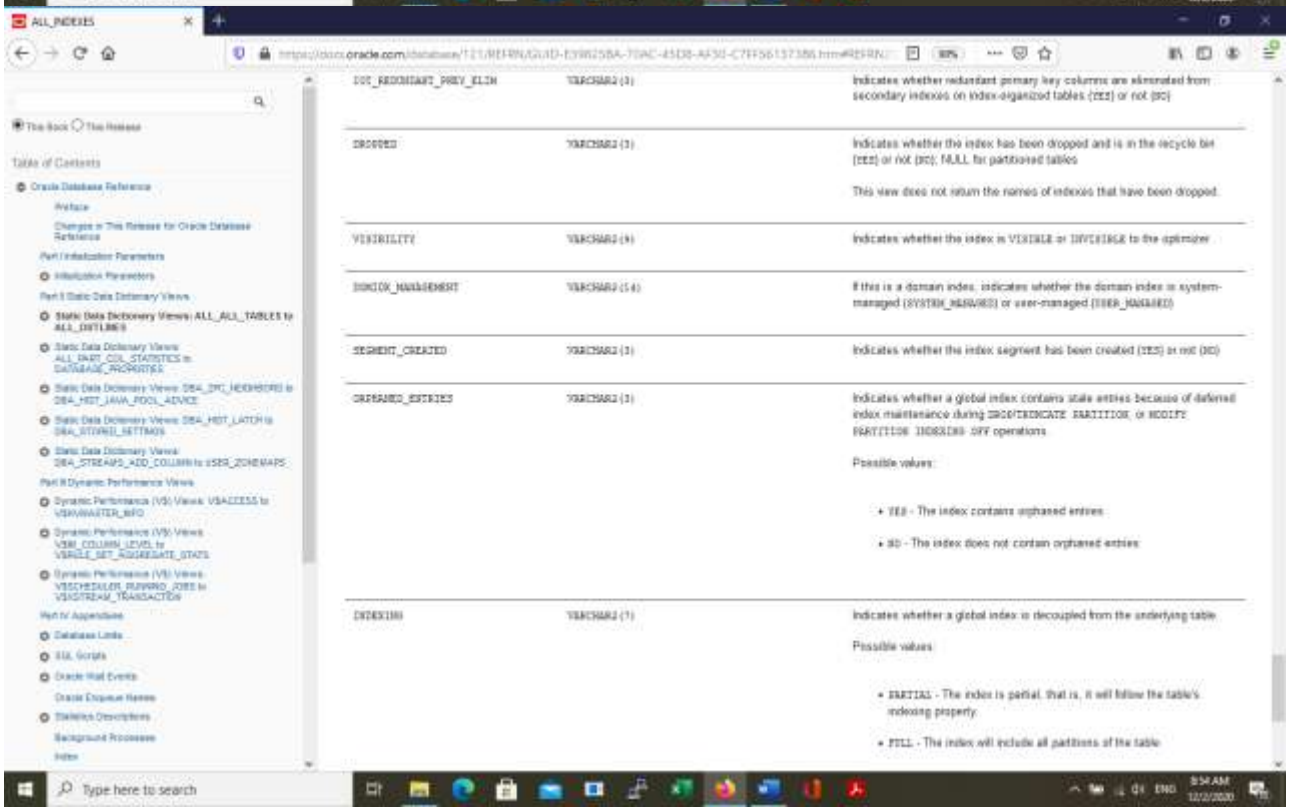
TABLE_OWNER	VARCHAR2 (128)	NOT NULL	Owner of the indexed object
TABLE_NAME	VARCHAR2 (128)	NOT NULL	Name of the indexed object
TABLE_TYPE	CHAR (11)		Type of the indexed object: • NEXT OBJECT • INDEX • TABLE • CLUSTER • VIEW • SYNONYM • SEQUENCE
UNIQUENESS	VARCHAR2 (9)		Indicates whether the index is unique (UNIQUE) or nonunique (NONUNIQUE)
COMPRESSION	VARCHAR2 (13)		The type of compression being used for the index: • ADVANCED LOW - Advanced low compression • DISABLED - No compression is present • ENABLED - Prefix compression
PREFIX_LENGTH	NUMBER		Number of columns in the prefix of the compression key



TABLESPACE_NAME	VARCHAR2 (30)		Name of the tablespace containing the index
INI_TRANS	NUMBER		Initial number of transactions
MAX_TRANS	NUMBER		Maximum number of transactions
INITIAL_EXTENT	NUMBER		Size of the initial extent
NEXT_EXTENT	NUMBER		Size of secondary extents
MIN_EXTENTS	NUMBER		Minimum number of extents allowed in the segment
MAX_EXTENTS	NUMBER		Maximum number of extents allowed in the segment
PCT_INCREASE	NUMBER		Percentage increase in extent size
PCT_THRESHOLD	NUMBER		Threshold percentage of block space allowed per index entry
INCLUDE_COLUMN	NUMBER		Column ID of the last column to be included in index-organized table primary key (non-overflow) index. This column maps to the COLUMN_ID column of the *_TAB_COLUMNS view
FREELISTS	NUMBER		Number of process freelists allocated to this segment
FREELIST_GROUPS	NUMBER		Number of freelist groups allocated to this segment
PCT_FREE	NUMBER		Minimum percentage of free space in a block
LOGGING	VARCHAR2 (3)		Indicates whether or not changes to the index are logged



STATUS	VARCHAR2 (8)	Indicates whether a nonpartitioned index is VALID or UNUSABLE
NUM_ROWS	NUMBER	Number of rows in the index For bitmap indexes, this column is the number of distinct keys, so its value is the same as the DISTINCT_KEYS column
SAMPLE_SIZE	NUMBER	Size of the sample used to analyze the index
LAST_ANALYZED	DATE	Date on which this index was most recently analyzed
DEGREE	VARCHAR2 (40)	Number of threads per instance for scanning the index, or DEFAULT
INSTANCES	VARCHAR2 (40)	Number of instances across which the indexes to be scanned, or DEFAULT
PARTITIONED	VARCHAR2 (3)	Indicates whether the index is partitioned (YES) or not (NO)
TEMPORARY	VARCHAR2 (1)	Indicates whether the index is on a temporary table (Y) or not (N)
GENERATED	VARCHAR2 (1)	Indicates whether the name of the index is system-generated (Y) or not (N)
SECONDARY	VARCHAR2 (1)	Indicates whether the index is a secondary object created by the ODCIIndexCreate method of the Oracle Data Cartridge (Y) or not (N)
BUFFER_POOL	VARCHAR2 (7)	Buffer pool to be used for index blocks • DEFAULT • KEEP • RECYCLE



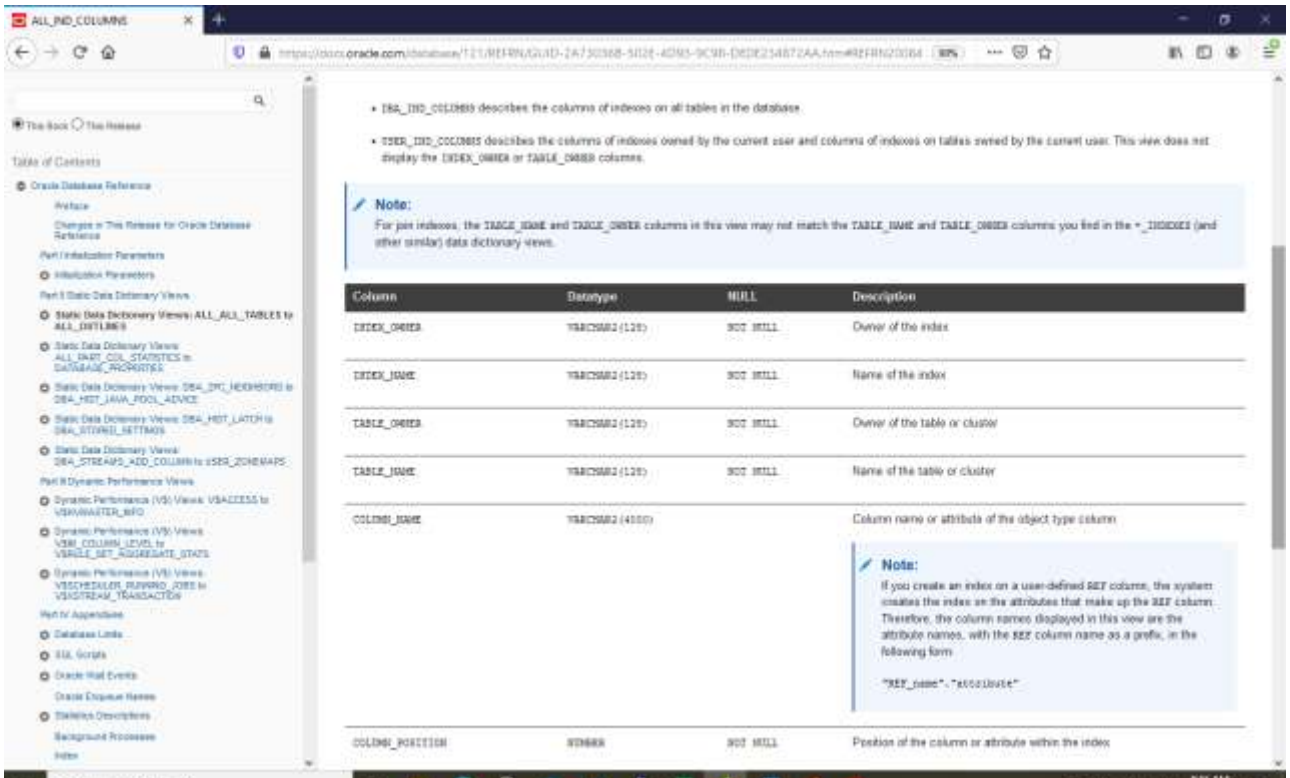
DUP_REDUNDANT_PREV_ELIM	VARCHAR2 (3)	Indicates whether redundant primary key columns are eliminated from secondary indexes on index-organized tables (YES) or not (NO)
DROPPED	VARCHAR2 (3)	Indicates whether the index has been dropped and is in the recycle bin (YES) or not (NO); NULL for partitioned tables This view does not return the names of indexes that have been dropped.
VISIBILITY	VARCHAR2 (9)	Indicates whether the index is VISIBLE or INVISIBLE to the optimizer
DOMIDX_MANAGEMENT	VARCHAR2 (14)	If this is a domain index, indicates whether the domain index is system-managed (SYSTEM_MANAGED) or user-managed (USER_MANAGED)
SEGMENT_CREATED	VARCHAR2 (3)	Indicates whether the index segment has been created (YES) or not (NO)
ORPHANED_ENTRIES	VARCHAR2 (3)	Indicates whether a global index contains stale entries because of deferred index maintenance during DROP/TRUNCATE PARTITION or MODIFY PARTITION INDEXING OFF operations Possible values: • YES - The index contains orphaned entries • NO - The index does not contain orphaned entries
INDEXING	VARCHAR2 (7)	Indicates whether a global index is decoupled from the underlying table Possible values: • PARTIAL - The index is partial, that is, it will follow the table's indexing property • FULL - The index will include all partitions of the table

```
SQL>  select  index_name, index_type, table_name, status from dba_indexes
          where owner='SCOTT' ;

INDEX_NAME                     INDEX_TYPE                 TABLE_NAME                     STATUS
------------------------------ -------------------------- ------------------------------ -------
DECIZII_PRIM                   NORMAL                     DECIZII                        VALID
DEPT_NAME_IDX                  BITMAP                     DEPT                           VALID
EMP_NAME_IDX                   NORMAL                     EMP                            VALID
PK_DEPT                        NORMAL                     DEPT                           VALID
PK_EMP                         NORMAL                     EMP                            VALID
PK_FUN                         NORMAL                     FUNCTII1                       VALID
PK_INT                         NORMAL                     INTRARI_GESTIUNE               VALID
PK_STOC                        NORMAL                     STOCURI                        VALID
```

8) Informatii din dictionar despre coloanele indecsilor:

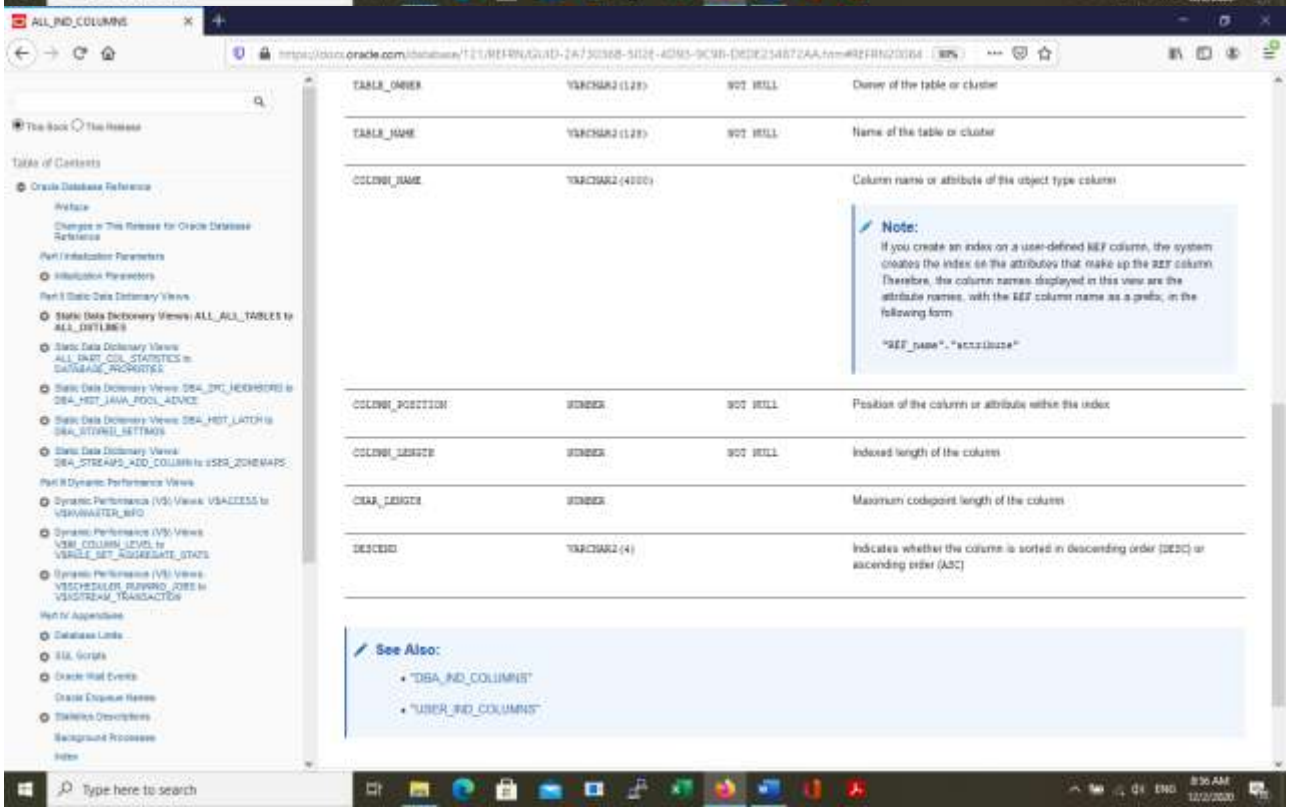
```
SQL> desc dba_ind_columns
 Name                           Null?                  Type
 ------------------------------ --------               ----
 INDEX_OWNER                    NOT NULL               VARCHAR2(30)
 INDEX_NAME                     NOT NULL               VARCHAR2(30)
 TABLE_OWNER                    NOT NULL               VARCHAR2(30)
 TABLE_NAME                     NOT NULL               VARCHAR2(30)
 COLUMN_NAME                                           VARCHAR2(4000)
 COLUMN_POSITION                NOT NULL               NUMBER
 COLUMN_LENGTH                  NOT NULL               NUMBER
 CHAR_LENGTH                                           NUMBER
 DESCEND                                               VARCHAR2(4)
```



- DBA_IND_COLUMNS describes the columns of indexes on all tables in the database.
- USER_IND_COLUMNS describes the columns of indexes owned by the current user and columns of indexes on tables owned by the current user. This view does not display the INDEX_OWNER or TABLE_OWNER columns.

Note:
For join indexes, the TABLE_NAME and TABLE_OWNER columns in this view may not match the TABLE_NAME and TABLE_OWNER columns you find in the *_INDEXES (and other similar) data dictionary views.

Column	Datatype	NULL	Description
INDEX_OWNER	VARCHAR2 (128)	NOT NULL	Owner of the index
INDEX_NAME	VARCHAR2 (128)	NOT NULL	Name of the index
TABLE_OWNER	VARCHAR2 (128)	NOT NULL	Owner of the table or cluster
TABLE_NAME	VARCHAR2 (128)	NOT NULL	Name of the table or cluster
COLUMN_NAME	VARCHAR2 (4000)		Column name or attribute of the object type column **Note:** If you create an index on a user-defined REF column, the system creates the index on the attributes that make up the REF column. Therefore, the column names displayed in this view are the attribute names, with the REF column name as a prefix, in the following form: "REF_name"."attribute"
COLUMN_POSITION	NUMBER	NOT NULL	Position of the column or attribute within the index



TABLE_OWNER	VARCHAR2 (128)	NOT NULL	Owner of the table or cluster
TABLE_NAME	VARCHAR2 (128)	NOT NULL	Name of the table or cluster
COLUMN_NAME	VARCHAR2 (4000)		Column name or attribute of the object type column **Note:** If you create an index on a user-defined REF column, the system creates the index on the attributes that make up the REF column. Therefore, the column names displayed in this view are the attribute names, with the REF column name as a prefix, in the following form: "REF_name"."attribute"
COLUMN_POSITION	NUMBER	NOT NULL	Position of the column or attribute within the index
COLUMN_LENGTH	NUMBER	NOT NULL	Indexed length of the column
CHAR_LENGTH	NUMBER		Maximum codepoint length of the column
DESCEND	VARCHAR2 (4)		Indicates whether the column is sorted in descending order (DESC) or ascending order (ASC)

See Also:

- "DBA_IND_COLUMNS"
- "USER_IND_COLUMNS"

```
SQL> select  index_name, table_owner, table_name, column_name
        from dba_ind_columns  where index_owner='SCOTT';

INDEX_NAME                     TABLE_OWNER                    TABLE_NAME                     COLUMN_NAME
------------------------------ ------------------------------ - ------------------------------ ----------------
DEPT_NAME_IDX                  SCOTT                          DEPT                             DNAME
EMPNAME_IDX                    SCOTT                          EMP                              ENAME
PK_COMP                        SCOTT                          COMPONENTE                        COD_COMP
PK_COMP                        SCOTT                           COMPONENTE                       PRET
PK_DEPT                        SCOTT                          DEPT                             DEPTNO
PK_EMP                         SCOTT                           EMP                             EMPNO
PK_INT                         SCOTT                          INTRARI_GESTIUNE                 NR_DOC_IN
PK_INT                         SCOTT                          INTRARI_GESTIUNE                 DATA_DOC_IN
PK_INT                         SCOTT                           INTRARI_GESTIUNE                COD_PRODUS
PK_INT                         SCOTT                          INTRARI_GESTIUNE                 COD_UM
PK_STOC                        SCOTT                           STOCURI                          COD_COMP
PK_STOC                        SCOTT                          STOCURI                          PRET
PK_STOC                        SCOTT                          STOCURI                          DATA_STOC
```

9) Startarea si stoparea monitorizarii unui index:

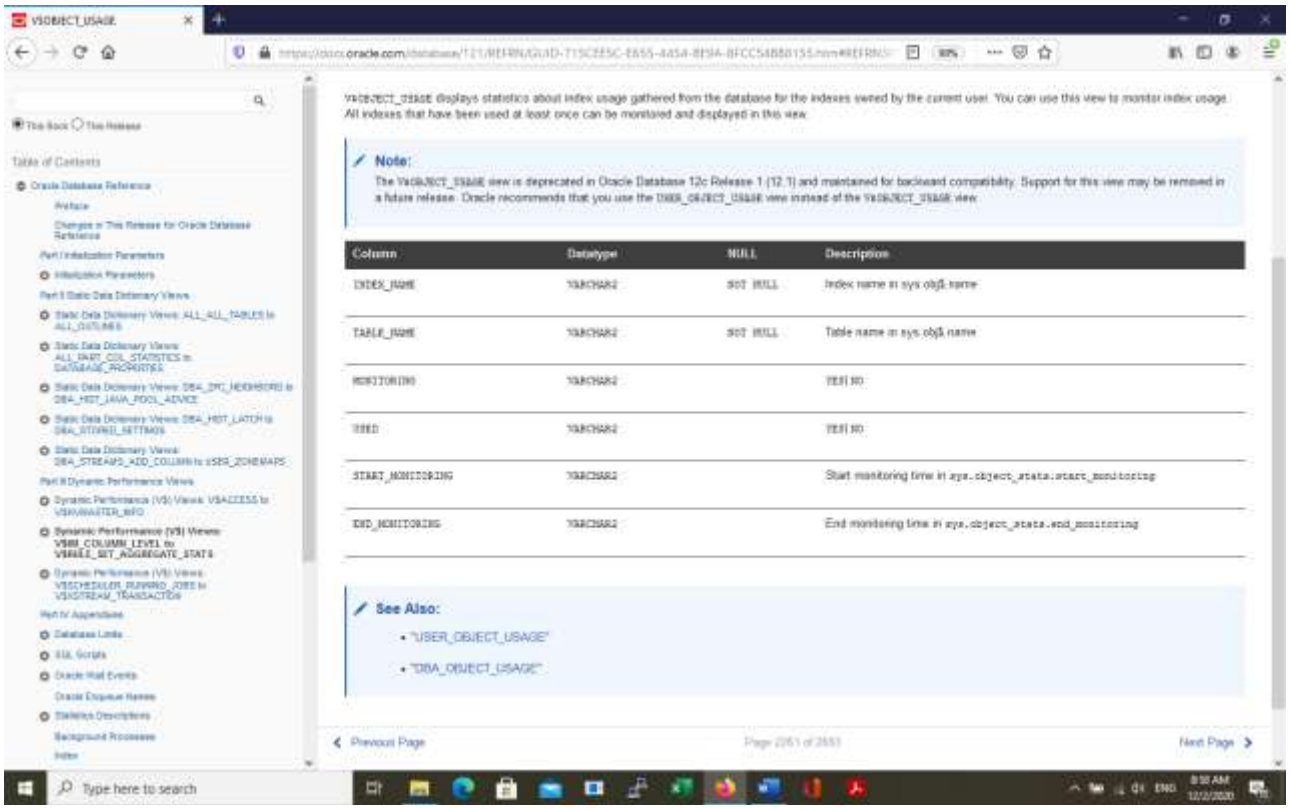
```
SQL> alter index emp_name_idx
        monitoring usage;
Index altered.

SQL> alter index emp_name_idx
        nomonitoring usage;
Index altered.
```

10) Informatii din dictionar despre indecsii monitorizati:

```
SQL> desc v$object_usage    (sau   user_object_usage)

 Name                           Null?              Type
 ------------------------------ --------           ----
 INDEX_NAME                      NOT NULL          VARCHAR2(30)
 TABLE_NAME                      NOT NULL          VARCHAR2(30)
 MONITORING                                        VARCHAR2(3)
 USED                                              VARCHAR2(3)
 START_MONITORING                                   VARCHAR2(19)
 END_MONITORING                                     VARCHAR2(19)
```



SQL> select * from v$object_usage;

INDEX_NAME	TABLE_NAME	MON	USE	START_MONITORING	END_MONITORING
EMP_NAME_IDX	EMP	NO	NO	11/15/2019 15:38:30	11/15/2019 15:41:26

11) Startarea analizei structurii unui index (se populeaza view-ul INDEX_STATS cu informatii despre index):

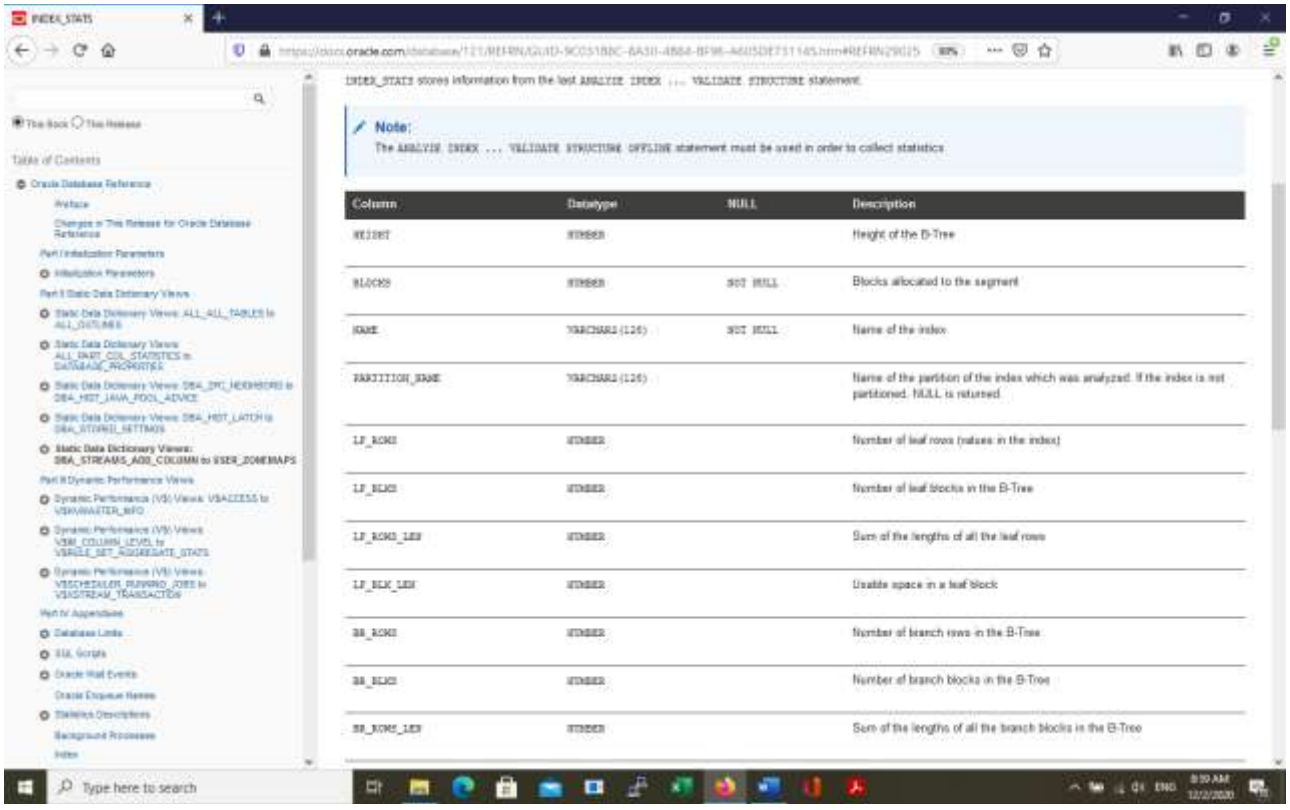
SQL> analyze index emp_name_idx validate structure;
Index analyzed.

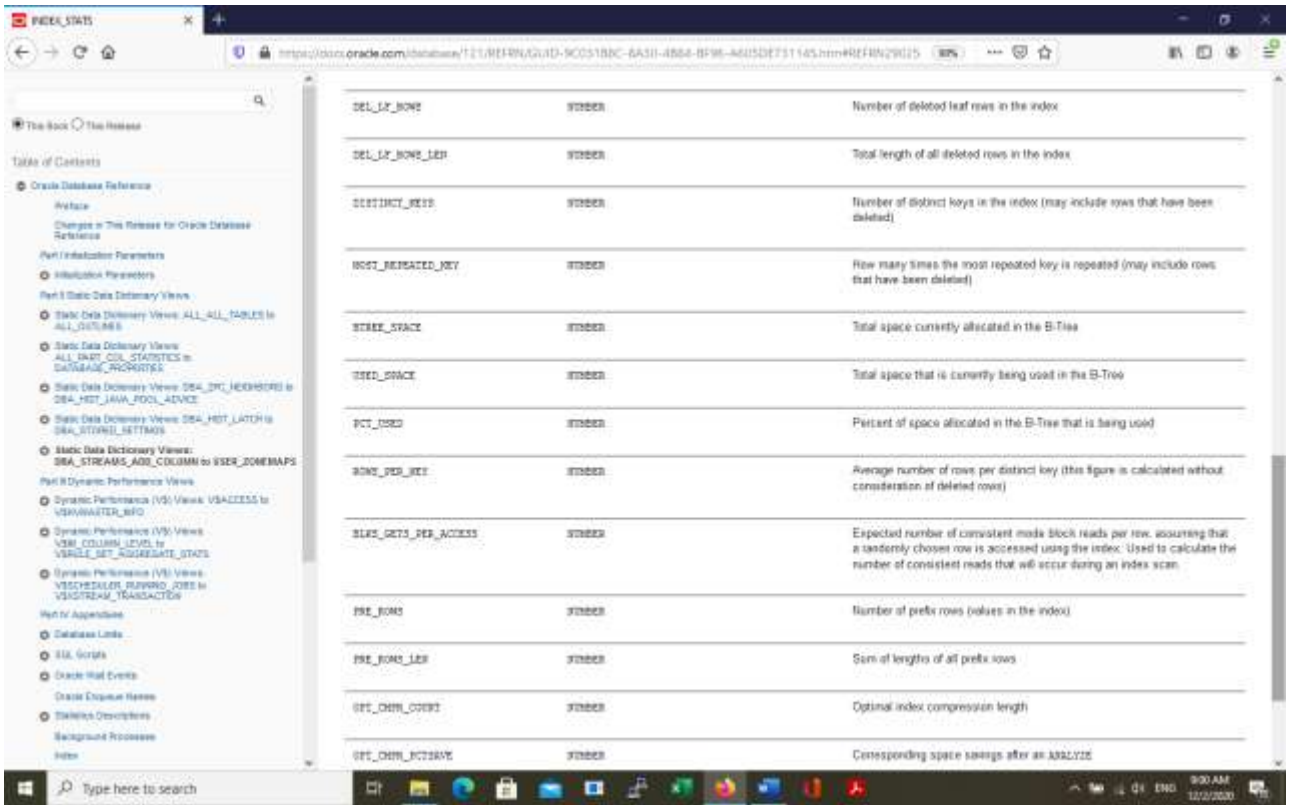
12) Informatii din dictionar despre starea indecsilor:

SQL> desc index_stats

Name	Null?	Type
HEIGHT		NUMBER
BLOCKS		NUMBER
NAME		VARCHAR2(30)
PARTITION_NAME		VARCHAR2(30)
LF_ROWS		NUMBER
LF_BLKS		NUMBER

LF_ROWS_LEN	NUMBER
LF_BLK_LEN	NUMBER
BR_ROWS	NUMBER
BR_BLKS	NUMBER
BR_ROWS_LEN	NUMBER
BR_BLK_LEN	NUMBER
DEL_LF_ROWS	NUMBER
DEL_LF_ROWS_LEN	NUMBER
DISTINCT_KEYS	NUMBER
MOST_REPEATED_KEY	NUMBER
BTREE_SPACE	NUMBER
USED_SPACE	NUMBER
PCT_USED	NUMBER
ROWS_PER_KEY	NUMBER
BLKS_GETS_PER_ACCESS	NUMBER
PRE_ROWS	NUMBER
PRE_ROWS_LEN	NUMBER
OPT_CMPR_COUNT	NUMBER
OPT_CMPR_PCTSAVE	NUMBER





SQL> select name, blocks, used_space, pct_used, distinct_keys, lf_rows, del_lf_rows
from index_stats ;

NAME	BLOCKS	USED_SPACE	PCT_USED	DISTINCT_KEYS	LF_ROWS	DEL_LF_ROWS
EMP_NAME_IDX	32	409	6	23	23	0

13) Stergerea unui index din dictionar:

SQL> drop index emp_name_idx ;
Index dropped.

Obs. Daca indexul este create printr-un constraint(de ex. unique sau cheie primara) nu se poate sterge direct, trebuie sters constraint-ul si se sterge automat si indexul:

SQL> ALTER TABLE table_name DROP CONSTRAINT constraint_name;